Communication-efficient ADMM over Hierarchical Networks *(Full Version)*

Binh Nguyen, Shuanqing Wei and Truong X. Nghiem

***Abstract*—This paper develops a novel distributed optimization algorithm based on the Alternating Direction Method of Multipliers (ADMM) to solve hierarchical optimization problems over tree-structured networks, termed hierarchical ADMM (hADMM), with a particular focus on enhancing communication efficiency across the network. By rearranging the augmented Lagrangian to establish a query-response communication mechanism between nodes that explicitly exploits the hierarchical tree structure, the proposed algorithm significantly reduces communication costs compared to existing ADMM-based methods for hierarchical optimization. Furthermore, hADMM guarantees asymptotic convergence under convexity assumptions. We also present a convergence rate analysis based on linear matrix inequalities to characterize the maximum theoretically achievable convergence rates across different network topologies, showing that the proposed hADMM attains linear convergence under mild conditions. Three numerical experiments demonstrate that hADMM is compatible with arbitrary tree network structures and outperforms existing approaches in terms of communication efficiency.**

***Index Terms*—ADMM, hierarchical optimization, sharing problem, tree network**

I. Introduction

The alternating direction method of multipliers (ADMM) is a robust optimization technique extensively employed in solving large-scale, distributed optimization problems characterized by structured objective functions and coupled constraints. ADMM integrates dual decomposition strategies with augmented Lagrangian methods, effectively breaking down complex optimization tasks into manageable subproblems that can be solved independently and concurrently. The method iteratively updates primal and dual variables, allowing distributed agents to solve local optimization tasks and subsequently coordinate solutions through dual variables. Due to its efficient convergence properties, ease of parallelization, and scalability, ADMM has seen wide adoption across numerous fields, including signal processing [1], machine learning [2], smart grid management [3], and robotics [4].

This work has been submitted to the IEEE for possible publication. Copyright may be transferred without notice, after which this version may no longer be accessible.

This material is based upon work supported by the National Science Foundation under Awards No. 2449927 and 231711.

Corresponding author: Truong X. Nghiem (truong.nghiem@ucf.edu).

Binh Nguyen and Truong X. Nghiem are with Department of Electrical and Computer Engineering, College of Engineering and Computer Science, University of Central Florida, Orlando, FL 32816, USA (e-mail: thanhbinh.nguyen@ucf.edu; truong.nghiem@ucf.edu).

Shuanqing Wei is with Division of Electrical and Computer Engineering, School of Electrical Engineering and Computer Science, Louisiana State University, Baton Rouge, LA 70803, USA (e-mail: swei@lsu.edu).

Distributed optimization over hierarchical networks (or simply hierarchical optimization) has attracted increasing attention due to its relevance in large-scale systems where agents are naturally organized in multi-level topologies. Such structures commonly arise in many applications [5], [6], [7], [8]. For example, in an electrical power distribution grid, each load, such as a building, is a leaf node with internal variables, dynamics, constraints, and total power demand determining its operating cost. Substations serving clusters of loads include transformers and equipment, and have their own dynamics, constraints, and aggregate demand based on connected loads. Each substation and its associated loads form a subtree. A central grid coordinator at the tree's root maintains grid stability and safety while providing services such as demand response [5], [9], [10], by coordinating substations and loads. In smart grids, two-way communication between nodes and their parents supports real-time demand adjustment and coordination, thus minimizing costs, ensuring reliability, and enhancing grid service performance. This setup results in a hierarchical optimization problem solvable in a distributed fashion.

Unlike flat peer-to-peer networks, hierarchical networks impose parent–child relationships and asymmetric information flows, which introduce additional challenges for distributed optimization, particularly in terms of coordination, scalability, and communication efficiency. Classical distributed optimization methods designed for fully connected or general graphs, e.g., [11], [12], [13], [14], [15], often *fail to fully exploit the structural properties*, leading to unnecessary communication overhead or reliance on centralized coordination. In particular, general hierarchical optimization problems can be solved by two simple ADMM-based methods. The first method, which we call the nested ADMM (nADMM) and was presented in [16], [17], solves a hierarchical sharing problem by applying classical ADMMs recursively to each layer of the tree, where each subtree is regarded by the ADMM applied at its parent as a single node for optimization purposes. The algorithm therefore involves multiple nested ADMM loops, one for each level of the tree. The second method, called the flattened ADMM (fADMM), flattens a tree network into a star network by establishing relay communication between each node and the root node. Conventional ADMM is then applied to the resulting star network, treating the original hierarchical structure as an equivalent star network. Both methods incur significant communication overhead, as we will show later in the experiments. Specifically, nADMM imposes heavy communication and computation loads on the deepest nodes, whereas fADMM places excessive relay communica-

tion demands on intermediate nodes.

Due to the above challenges, there is a growing need for *structure-aware distributed optimization* frameworks that explicitly leverage the hierarchical topology to enable scalable, communication-efficient, and fully distributed computation. This paper presents a novel ADMM-based algorithm, called the hierarchical ADMM (hADMM), that effectively exploits the hierarchical structure to drastically improve communication efficiency. Our **main contributions** are summarized as follow:

- We propose a novel hADMM algorithm for efficiently solving general hierarchical optimization problems over arbitrary tree-structured networks, achieving substantial reductions in network communication costs in comparison with the fADMM and the nADMM.
- We provide rigorous theoretical analyses establishing the convergence and convergence rate of the proposed method. In particular, the hADMM is guaranteed to converge under general convexity assumptions, even in the presence of nonsmooth convex cost functions.
- We demonstrate the superior communication efficiency of the hADMM compared with the fADMM and the nADMM and validate the theoretical results through comprehensive and systematic numerical experiments.

Notation: Let $\mathbb{R}$, $\mathbb{R}_{>0}$, $\mathbb{N}$, and $\mathbb{N}_{>0}$ be the sets of real numbers, positive real numbers, integers, and positive integers, respectively. Denote $\partial f(x)$ as the subdifferential of a function f at x. For $D \in \mathbb{N}_{>0}$, define $\mathbb{N}_D = \{1, \ldots, D\}$. In symmetric block matrices, the asterisk $(*)$ is used as an ellipsis for terms induced by symmetry. For integers $n, m \in \mathbb{N}_{>0}$, $I_n \in \mathbb{R}^{n\times n}$ denotes the identity matrix of size n, 0_n is the n-dimensional zero vector, and $0_{n\times m}$ is the $n\times m$ zero matrix. For an ordered index list $\mathcal{I} = \{i_1, i_2, \ldots, i_n\}$, $i_k \in \mathbb{N}_{>0}$, and real matrices $M_{i_1}, M_{i_2}, \ldots, M_{i_n}$ in appropriate dimensions, let $\{M_i\}_{i\in\mathcal{I}}$ be the ordered list of the matrices, and denote

$$[M_i]_{i\in\mathcal{I}} = \left[M_{i_1}^\top, M_{i_2}^\top, \ldots, M_{i_n}^\top\right]^\top,$$

$$[M_i]_{i\in\mathcal{I}}^{\text{diag}} = \text{diag}(M_{i_1}, \ldots, M_{i_n}) = \begin{bmatrix} M_{i_1} & & \\ & \ddots & \\ & & M_{i_n} \end{bmatrix}.$$

For real matrices $U_1, \ldots, U_{n-1}$, $D_1, \ldots, D_n$, and $L_1, \ldots, L_{n-1}$ in appropriate dimensions, denote the tridiagonal block matrix

$$\text{tridiag}\left(\{U_i\}_{i\in\mathbb{N}_{n-1}}, \{D_i\}_{i\in\mathbb{N}_n}, \{L_i\}_{i\in\mathbb{N}_{n-1}}\right) = \begin{bmatrix} D_1 & U_1 & \ldots & 0 \\ L_1 & D_2 & \ddots & \vdots \\ \vdots & \ddots & \ddots & U_{n-1} \\ 0 & \ldots & L_{n-1} & D_n \end{bmatrix}.$$

For a set $\mathcal{S}$, $|\mathcal{S}|$ is the number of elements in $\mathcal{S}$. For a real matrix M, $\|M\|_2$ denotes the matrix 2-norm (spectral norm) of M, which is the largest singular value of M. For two square matrices A and B of the same dimensions, the notation $A \succ (\succeq) B$ implies that $A - B$ is a positive (semi)definite matrix.

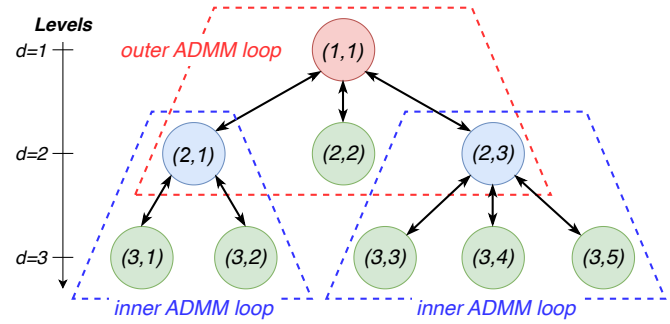


Fig. 1. An example of bidirectional tree networks: the root node, parent nodes, and leaf nodes are red, blue, and green, respectively. It also illustrates the nested ADMM scheme.

II. Problem Formulation and Related Literature

This section formulates a hierarchical optimization problem defined over a tree network. We then present and discuss the nested and flattened ADMM schemes, which serve as baseline approaches for the comparative evaluation of our proposed methods.

A. Hierarchical optimization over tree networks

Let us consider a bidirectional tree network $\mathcal{G}$ with $D \in \mathbb{N}_{>0}$ levels (layers). A pair (d, i) stands for the i-th node at level d, where $d = 1$ is the root level and $(1, 1)$ is the one and only root node. A parent node is a node at level $d \in \{1, \ldots, D-1\}$ which is connected to at least a node at the next level $d+1$, called a child node. A node that is not a parent node is called a leaf node. Each non-root node has exactly one parent node and may have multiple child nodes. A node at level d can communicate with only its parent node and its child nodes through their direct connections, hence it is $(d-1)$ hops from the root node in terms of communication.

Let $\mathcal{V} \subset \mathbb{N}_{>0}^2$ be the set of all nodes. For node $(d, i) \in \mathcal{V}$, denote $\mathcal{C}_{d,i} = \{j \in \mathbb{N}_{>0} | (d+1, j) \in \mathcal{V} \text{ is a child of } (d, i)\}$ as the set of its children. Let $\mathcal{P}_d = \{i \in \mathbb{N}_{>0} | (d, i) \in \mathcal{V}, \mathcal{C}_{d,i} \neq \emptyset\}$ be the set of all parent nodes at level d and $\mathcal{L}_d = \{i \in \mathbb{N}_{>0} | (d, i) \in \mathcal{V}, \mathcal{C}_{d,i} = \emptyset\}$ be the set of all leaf nodes at level d. Note that $\mathcal{P}_d \cup \mathcal{L}_d$ contains the indices of all nodes at level d. For example, consider the tree network with three levels ($D = 3$) in Fig. 1. The first layer includes the root node denoted by $(1, 1)$ with $\mathcal{P}_1 = \{1\}$, $\mathcal{C}_{1,1} = \{1, 2, 3\}$, and $\mathcal{L}_1 = \emptyset$. The second layer includes three nodes $(2, 1), (2, 2)$, and $(2, 3)$, with $\mathcal{P}_2 = \{1, 3\}$, $\mathcal{L}_2 = \{2\}$, $\mathcal{C}_{2,1} = \{1, 2\}$, $\mathcal{C}_{2,2} = \emptyset$, and $\mathcal{C}_{2,3} = \{3, 4, 5\}$. The third layer has only leaf nodes, with $\mathcal{P}_3 = \emptyset$ and $\mathcal{L}_3 = \{1, 2, 3, 4, 5\}$.

Each node (d, i) has a vector of decision variables $x_{d,i} \in \mathbb{R}^{n_{d,i}}$ and a cost function $f_{d,i}(x_{d,i})$. This paper considers the following *hierarchical optimization* problem over $\mathcal{G}$:

$$\text{minimize} \sum_{d=1}^{D} \left(\sum_{i\in\mathcal{P}_d} f_{d,i}(x_{d,i}) + \sum_{j\in\mathcal{L}_d} f_{d,j}(x_{d,j}) \right), \quad \text{(1a)}$$

$$\text{s.t. } A_{d,ij} x_{d,i} + B_{d,ij} x_{d+1,j} = c_{d,ij}, \quad \text{(1b)}$$

where (1b) describes the relationship between node $(d+1, j)$ and its parent (d, i) and holds for all $d \in \mathbb{N}_{D-1}$, $i \in \mathcal{P}_d$,

and $j \in \mathcal{C}_{d,i}$, with matrices $A_{d,ij} \in \mathbb{R}^{s_{d,ij} \times n_{d,i}}$, $B_{d,ij} \in \mathbb{R}^{s_{d,ij} \times n_{d+1,j}}$, and vector $c_{d,ij} \in \mathbb{R}^{s_{d,ij}}$. It is assumed that $f_{d,i}(x_{d,i})$ are convex functions for all $(d,i) \in \mathcal{V}$, thus the optimization problem (1) is convex.

B. Disjoint problems over tree networks

The *disjoint problem* is a special case of the hierarchical optimization problem (1), that is often encountered in practice. In this problem, the decision vector of a parent node at level d is the concatenation of the decision vectors of its children, that is, $x_{d,i} = [x_{d+1,j}]_{j \in \mathcal{C}_{d,i}}$. The resulting problem

$$\text{(1a) s.t. } x_{d,i} = [x_{d+1,j}]_{j \in \mathcal{C}_{d,i}},\ \forall d \in \mathbb{N}_{D-1}, i \in \mathcal{P}_d \quad (2)$$

can be formulated as the general problem (1) with $n_{d,i} = \sum_{j \in \mathcal{C}_{d,i}} n_{d+1,j}$, $c_{d,ij} = 0$, $B_{d,ij} = -I_{n_{d+1,j}}$, and $A_{d,ij} = \begin{bmatrix} 0 & \dots & I_{n_{d+1,j}} & \dots & 0 \end{bmatrix} \in \mathbb{R}^{n_{d+1,j} \times n_{d,i}}$ where the block of $A_{d,ij}$ corresponding to the child node $j \in \mathcal{C}_{d,i}$ is the identity matrix and all the other blocks are zero. Here, $A_{d,ij}$ effectively selects the variables $x_{d+1,j}$ corresponding to the child node j in the vector $x_{d,i}$. Note that $[A_{d,ij}]_{j \in \mathcal{C}_{d,i}} = I_{n_{d,i}}$.

The rest of this section describes two standard algorithms for solving the general hierarchical optimization problem (1).

C. Nested ADMM (nADMM) algorithm

The standard ADMM can be adapted to solve (1). Specifically, ADMM solves the two-block coupled problem [18]

$$\underset{u,v}{\text{minimize}}\ g(u) + h(v) \text{ s.t. } Au + Bv = c, \quad (3)$$

with convex functions g and h, by using the following uncoupled steps, for $\rho > 0$,

$$\begin{cases} u^{k+1} = \arg\min_u g(u) + \frac{\rho}{2}\|Au + Bv^k - c + \frac{1}{\rho}y^k\|_2^2, \\ v^{k+1} = \arg\min_v h(v) + \frac{\rho}{2}\|Au^{k+1} + Bv - c + \frac{1}{\rho}y^k\|_2^2, \\ y^{k+1} = y^k + \rho(Au^{k+1} + Bv^{k+1} - c). \end{cases} \quad (4)$$

As shown in [18], the iterations (4) ensure the stable sequence (u^k, v^k, y^k), which converges to the optimal solution when g and h are closed proper convex functions, and A or B is a full rank matrix [19]. The authors in [20] show that the ADMM (4) converges in the dual objective with linear rate, and they further propose an accelerated variant of (4) Other results on linear convergence of the ADMM are shown in [21]. A more general variant of (3), called the multiple-block problem,

$$\underset{u,v_1,\dots,v_N}{\text{minimize}}\ g(u) + \sum_{i=1}^{N} h_i(v_i) \text{ s.t. } Au + \sum_{i=1}^{N} B_i v_i = c, \quad (5)$$

has been solved by Jacobi-type ADMM [22] and proximal Jacobi-type ADMM [23] with linear convergence guarantee.

Both (3) and its generalization (5) are formulated for star networks, where g is the cost function of the coordinator and h_i is the local cost function of the leaf node i. For solving the hierarchical optimization (1), the authors of [16], [17], [24] apply standard ADMM (4) recursively to each layer of the tree structure. We term this method *nested ADMM* (nADMM). Other variants of nADMM can be found in [25], [26], which address federated learning over tree networks. In nADMM, each subtree is regarded by the ADMM applied at its parent node as a single node for optimization purposes. The algorithm therefore involves multiple nested loops, one for each level of the tree. Consider the illustrative example in Fig. 1. The root node $(1,1)$, acting as the coordinator, uses the ADMM to solve the global optimization problem. In each iteration of the ADMM, when a cluster node at depth 2, such as $(2,1)$ or $(2,3)$, is queried, it applies the ADMM to solve its local proximal minimization corresponding to its subtree. This is illustrated in Fig. 1, where the blue boxes represent the inner-level ADMM loops applied locally to parent nodes $(2,1)$ and $(2,3)$, and the red box represents the outer-level ADMM loop at the root node. A query from the root node to a cluster node $(2,1)$ or $(2,3)$ results in complete inner-level ADMM loops for the cluster node with respect to its subtree, involving multiple inner query-response rounds, before a response is sent back to the root node.

While the algorithms in [16] and [17] are designed only for the disjoint problem (2) over tree networks with depth $D = 3$, we generalize the nADMM for the hierarchical optimization problem (1) over a general tree network in Appendix A.

D. Flattened ADMM (fADMM) algorithm

In the flattened ADMM, the global objective function (1a) is first flattened by rewriting it as a single sum of the local objective functions of all individual nodes, then the standard ADMM is applied where all nodes are considered equal, i.e., each iteration involves solving the proximal minimization problem of every node in the tree. A non-root parent node will relay the data exchanges between its children and its parent. Therefore, from the computation perspective, the tree is flattened into a star structure. From the communication perspective, however, the tree structure is retained as the data flow follows the tree.

E. Communication and computation costs of nADMM and fADMM

The fADMM incurs significant communication overhead as messages must be exchanged between the root and every node in the network along a multi-hop path between them, where all nodes in the path must relay messages in both directions. Such a process is particularly costly in terms of communication when the non-root node resides at the deepest level of the tree [17]. Moreover, the dependence of fADMM on multi-hop communication increases network protocol complexity and generates additional traffic on the underlying infrastructure, which can compromise its overall reliability. The overall computation cost of the fADMM is proportional to the network size (i.e., the number of nodes) and the number of iterations.

In contrast to the fADMM, the nADMM only requires direct (single-hop) communication between a parent node and a child node. However, this advantage comes at the cost of high computational complexity and does not reduce the overall communication cost. Specifically, each iteration of an outer loop involves recursive computations in the inner loops, leading to an exponential growth in communication and computation costs as the network depth increases. As

TABLE I
SUMMARY OF IMPORTANT NOTATIONS

Notation	Meaning
D	Depth of the tree network $\mathcal{G}$
(d,i)	i-th node at d-th level of $\mathcal{G}$
$\mathcal{C}_{d,i}$	Index set of all children of node (d,i)
$\mathcal{P}_d$	Index set of all parent nodes at level d
$\mathcal{L}_d$	Index set of all leaf nodes at level d
$x_{d,i}$	Decision variable vector of node (d,i)
$f_{d,i}$	Cost function of node (d,i)
x_d	Concatenation of all node variables at level d
$s_{d,ij}$	Number of scalar constraints between node (d,i) and its child $(d+1,j)$

a result, lower-level nodes perform more iterations than do upper-level nodes. Consequently, the root node incurs the least communication and computation costs, while nodes at the deepest level have the highest communication and computation burden. Overall, the total communication and computation costs of the nADMM are higher than those of the fADMM.

III. HIERARCHICAL ADMM (HADMM) ALGORITHM

We propose an algorithm, called *hierarchical ADMM* (hADMM), for solving the hierarchical optimization problem (1), where computation and communication naturally follow the tree structure of the network, resulting in computation and communication savings compared to the nADMM and fADMM algorithms. These savings will be demonstrated in the numerical examples in Section V. To improve readability of the mathematical derivation afterward, selected important notations are summarized in Table I.

A. Augmented Lagrangian function

This section establishes an augmented Lagrangian function for the problem (1). We first define some useful notations. The constraints (1b) for $j \in \mathcal{C}_{d,i}$ can be combined into an aggregate constraint between the parent node (d,i) and all its child nodes $(d+1,j)$ as $A_{d,i}x_{d,i}+B_{d,i}[x_{d+1,j}]_{j\in\mathcal{C}_{d,i}} = c_{d,i}$, where $A_{d,i} = [A_{d,ij}]_{j\in\mathcal{C}_{d,i}}$, $B_{d,i} = [B_{d,ij}]^{\text{diag}}_{j\in\mathcal{C}_{d,i}}$, and $c_{d,i} = \left[c_{d,ij}\right]_{j\in\mathcal{C}_{d,i}}$. Let us define the vector of all node variables at level d as $x_d = \left[[x_{d,i}]^\top_{i\in\mathcal{P}_d}, [x_{d,i}]^\top_{i\in\mathcal{L}_d}\right]^\top \in \mathbb{R}^{n_d}$ and the sum of all local cost functions at level d as $f_d(x_d) = \sum_{i\in\mathcal{P}_d\cup\mathcal{L}_d} f_{d,i}(x_{d,i})$. The hierarchical optimization problem (1) can be expressed as the following multiple-layer optimization problem:

$$\underset{x_1,\dots,x_D}{\text{minimize}}\ J = \sum_{d=1}^{D} f_d(x_d), \tag{6a}$$

$$\text{s.t. } A_d x_d + B_d x_{d+1} = c_d,\ \forall d \in \mathbb{N}_{D-1}, \tag{6b}$$

where $A_d = \left[[A_{d,i}]^{\text{diag}}_{i\in\mathcal{P}_d},\ 0\right] \in \mathbb{R}^{s_d\times n_d}$ with $s_d = \sum_{i\in\mathcal{P}_d}\sum_{j\in\mathcal{C}_{d,i}} s_{d,ij}$, $B_d = [B_{d,i}]^{\text{diag}}_{i\in\mathcal{P}_d} M_d$, $c_d = \left[c_{d,i}\right]_{i\in\mathcal{P}_d}$, and M_d is a permutation matrix for extracting the variables of all the child nodes at level $d+1$, i.e., $[[x_{d+1,j}]_{j\in\mathcal{C}_{d,i}}]_{i\in\mathcal{P}_d} = M_d x_{d+1}$. For example, in Fig. 1, at $d=1$ we have

$$x_1 = x_{1,1},\ x_2 = \begin{bmatrix} x_{2,1} \\ x_{2,3} \\ x_{2,2} \end{bmatrix},\ M_1 = \begin{bmatrix} I_{n_{2,1}} & 0 & 0 \\ 0 & 0 & I_{n_{2,3}} \\ 0 & I_{n_{2,2}} & 0 \end{bmatrix}.$$

We establish the following augmented Lagrangian for optimization problem (6) with penalty parameter $\rho > 0$:

$$L = f_D(x_D) + \sum_{d=1}^{D-1}\Big(f_d(x_d) + y_d^\top(A_d x_d + B_d x_{d+1} - c_d) + \frac{\rho}{2}\|A_d x_d + B_d x_{d+1} - c_d\|_2^2\Big), \tag{7}$$

where $y_d = [y_{d,i}]_{i\in\mathcal{P}_d} \in \mathbb{R}^{s_d}$, $y_{d,i} = [y_{d,ij}]_{j\in\mathcal{C}_{d,i}}$, and $y_{d,ij}$ is the dual variable associated with the constraint (1b) between the node (d,i) and its child node $(d+1,j)$. The Lagrangian can be rearranged as a sum of multiple Lagrangians for each layer d of the tree network as:

$$L = \sum_{d=1}^{D}\left(L_d^{\text{p}}(x_d, x_{d+1}, y_d) + L_d^{\text{e}}(x_d)\right), \tag{8}$$

where $L_D^{\text{p}} = 0$, $L_1^{\text{e}} = 0$, $L_d^{\text{p}}(x_d, x_{d+1}, y_d) = \sum_{i\in\mathcal{P}_d}\Big(f_{d,i}(x_{d,i}) + y_{d,i}^\top(A_{d,i}x_{d,i} + B_{d,i}[x_{d+1,j}]_{j\in\mathcal{C}_{d,i}} - c_{d,i}) + \frac{\rho}{2}\|A_{d,i}x_{d,i} + B_{d,i}[x_{d+1,j}]_{j\in\mathcal{C}_{d,i}} - c_{d,i}\|_2^2\Big)$ for $1 \le d \le D-1$, and $L_d^{\text{e}}(x_d) = \sum_{i\in\mathcal{L}_d} f_{d,i}(x_{d,i})$ for $2 \le d \le D$.

B. Hierarchical ADMM (hADMM) algorithm

The *key idea* of our proposed hADMM algorithm is to hierarchically solve primal variables top-down from the root node to the leaf nodes, then update the dual variables bottom-up from the leaf nodes to the root node. Specifically, inpsired by the Jacobian-type ADMM [22], [27], we formulate the hADMM algorithm for the hierarchical optimization problem (1) into a forward pass and a backward pass as follows.

Forward:

$$\begin{aligned} x_1^{k+1} &= \arg\min_{x_1} L(x_1, x_2^k, \dots, x_D^k; y_1^k, \dots) \\ x_2^{k+1} &= \arg\min_{x_2} L(x_1^{k+1}, x_2, x_3^k, \dots, x_D^k; y_1^k, \dots) \\ &\dots \\ x_D^{k+1} &= \arg\min_{x_D} L(x_1^{k+1}, \dots, x_{D-1}^{k+1}, x_D; y_1^k, \dots) \end{aligned} \tag{9}$$

Backward:

$$\begin{aligned} y_{D-1}^{k+1} &= y_{D-1}^k + \rho(A_{D-1}x_{D-1}^{k+1} + B_{D-1}x_D^{k+1} - c_{D-1}) \\ y_{D-2}^{k+1} &= y_{D-2}^k + \rho(A_{D-2}x_{D-2}^{k+1} + B_{D-2}x_{D-1}^{k+1} - c_{D-2}) \\ &\dots \\ y_1^{k+1} &= y_1^k + \rho(A_1 x_1^{k+1} + B_1 x_2^{k+1} - c_1) \end{aligned} \tag{10}$$

It is noted that the computation and communication between nodes in the forward and backward passes are distributed and follow the tree structure. Indeed, let us consider any node (d,i). The forward pass (9) results in three cases as follows.
(i) If $(d=1, i=1)$ is the root node, where $x_1 = x_{1,1}$:

$$\begin{aligned} x_{1,1}^{k+1} &= \arg\min_{x_{1,1}} L_1^{\text{p}}(x_1, x_2^k, y_1^k) \\ &= \arg\min_{x_{1,1}} f_{1,1}(x_{1,1}) + y_{1,1}^{k\top}\Big(A_1 x_1 + B_1 x_2^k - c_1\Big) \\ &\qquad + \frac{\rho}{2}\|A_1 x_1 + B_1 x_2^k - c_1\|_2^2. \end{aligned} \tag{11}$$

Algorithm 1 Distributed implementation of the hADMM for solving the hierarchical optimization problem (1).

function hADMM$((d,i),q)$
  if (d,i) is leaf node **then**
    Compute $x_{d,i}$ by (12)
    Send $x_{d,i}$ to parent $(d-1,p)$
    return
  end if
  if $(d,i)=(1,1)$ **then** ▷ root node
    Compute $x_{1,1}$ by (11)
  else
    Compute $x_{d,i}$ by (13)
    Send $x_{d,i}$ to parent $(d-1,p)$
  end if
  for $j\in\mathcal{C}_{d,i}$ **do** ▷ in parallel
    $q_{d,ij}=A_{d,ij}x_{d,i}+\frac{1}{\rho}y_{d,ij}-c_{d,ij}$
    Send $q_{d,ij}$ to $(d+1,j)$ ▷ to child $(d+1,j)$
    $x_{d+1,j}=$hADMM$((d+1,j),q_{d,ij})$ ▷ from child
    $r_{d,ij}=A_{d,ij}x_{d,i}+B_{d,ij}x_{d+1,j}-c_{d,ij}$
    Update $y_{d,ij}\leftarrow y_{d,ij}+\rho r_{d,ij}$
  end for
end function
repeat
  hADMM$((1,1),[])$ ▷ Solve (1) at root node
until convergence as in Section III-C

(ii) If (d,i) is a leaf node with the parent node $(d-1,p)$:

$$\begin{aligned}x_{d,i}^{k+1}&=\arg\min_{x_{d,i}} L_{d-1}^{\mathrm{p}}(x_{d-1}^{k+1},x_d,y_{d-1}^k)+L_d^{\mathrm{e}}(x_d)\\&=\arg\min_{x_{d,i}} f_{d,i}(x_{d,i})\\&\quad+(y_{d-1,pi}^k)^\top(A_{d-1,pi}x_{d-1,p}^{k+1}+B_{d-1,pi}x_{d,i}-c_{d-1,pi})\\&\quad+\frac{\rho}{2}\|A_{d-1,pi}x_{d-1,p}^{k+1}+B_{d-1,pi}x_{d,i}-c_{d-1,pi}\|_2^2.\end{aligned}\tag{12}$$

(iii) If (d,i) is a non-leaf node with the parent node $(d-1,p)$:

$$\begin{aligned}x_{d,i}^{k+1}&=\arg\min_{x_{d,i}} L_{d-1}^{\mathrm{p}}(x_{d-1}^{k+1},x_d,y_{d-1}^k)+L_d^{\mathrm{p}}(x_d,x_{d+1}^k,y_d^k)\\&=\arg\min_{x_{d,i}} f_{d,i}(x_{d,i})\\&\quad+\sum_{j\in\mathcal{C}_{d,i}}\Big((y_{d,ij}^k)^\top(A_{d,ij}x_{d,i}+B_{d,ij}x_{d+1,j}^k-c_{d,ij})\\&\quad+\frac{\rho}{2}\|A_{d,ij}x_{d,i}+B_{d,ij}x_{d+1,j}^k-c_{d,ij}\|_2^2\Big)\\&\quad+(y_{d-1,pi}^k)^\top(A_{d-1,pi}x_{d-1,p}^{k+1}+B_{d-1,pi}x_{d,i}-c_{d-1,pi})\\&\quad+\frac{\rho}{2}\|A_{d-1,pi}x_{d-1,p}^{k+1}+B_{d-1,pi}x_{d,i}-c_{d-1,pi}\|_2^2.\end{aligned}\tag{13}$$

Based on these three cases, we summarize the hADMM algorithm in Algorithm 1, which is a distributed implementation of the forward and backward passes (9) and (10).

C. Stopping criterion

For each non-leaf node (d,i), we compute the maximum primal residual $\max_{j\in\mathcal{C}_{d,i}}\|r_{d,ij}^{k+1}\|_\infty$ of all children, where $r_{d,ij}^{k+1}=A_{d,ij}x_{d,i}^{k+1}+B_{d,ij}x_{d+1,j}^{k+1}-c_{d,ij}^{k+1}$. The stopping criterion of the hADMM is given by

$$\max_{d\in\mathbb{N}_{D-1},i\in\mathcal{P}_d}\ \max_{j\in\mathcal{C}_{d,i}}\|r_{d,ij}^{k+1}\|_\infty\le\varepsilon,\tag{14}$$

where $\varepsilon>0$ is a predefined tolerance. Note that this stopping criterion can be checked in a distributed manner as follows. Each non-leaf node (d,i) computes its local maximum primal residual $\max_{j\in\mathcal{C}_{d,i}}\|r_{d,ij}^{k+1}\|_\infty$ and sends it to its parent node $(d-1,p)$. The root node $(1,1)$ collects all the local maximum primal residuals from its child nodes and checks the stopping criterion (14).

TABLE II
SCENARIOS AND CONVERGENCE RESULTS OF THE HADMM.

Scenario	Theoretical results
1: convex f_d, full column rank B_d	Theorem 1: Convergence
2: convex f_1, differentiable strongly convex $f_2,\dots,f_D$	Theorem 2: Convergence
3: differentiable convex f_1, strongly convex $f_2,\dots,f_D$ with Lipschitz-continuous gradients	Theorem 3: Linear convergence rate

IV. Convergence Analysis of hADMM

This section provides theoretical guarantees for the convergence of the proposed hADMM in Algorithm 1 and a theoretical analysis of its convergence rate. We consider the following scenarios in our convergence analysis, based on [28], [23], [29], that are commonly assumed in the literature. The convergence analysis results are summarized in Table II.

Scenario 1. *Functions $f_1,f_2,\dots$ and f_D are closed proper convex, and matrices $B_1,\dots,B_{D-1}$ have full column rank.*

Scenario 2. *Function f_1 is closed proper convex, functions $f_2,\dots,f_D$ are differentiable and strongly convex with moduli $\alpha_d>0$, that is, for all $z,z'\in\mathrm{Dom}(f_d)$,*

$$f_d(z)-f_d(z')\ge\partial f_d^\top(z')(z-z')+\alpha_d\|z-z'\|_2^2.\tag{15}$$

A. Convergence of hADMM

We assume that there is an optimal primal solution $[x_d^\star]_{d\in\mathbb{N}_D}$ and an optimal dual solution $[y_d^\star]_{d\in\mathbb{N}_{D-1}}$ satisfying the generalized Karush-Kuhn-Tucker (KKT) condition for the optimization problem (1) as follow:

$$0\in\partial f_1(x_1^*)+A_1^\top y_1^*\tag{16a}$$
$$0\in\partial f_D(x_D^*)+B_{D-1}^\top y_{D-1}^*\tag{16b}$$
$$0\in\partial f_d(x_d^*)+A_d^\top y_d^*+B_{d-1}^\top y_{d-1}^*,\ d=2,\dots,D-1\tag{16c}$$
$$A_dx_d^*+B_dx_{d+1}^\star=c_d,\ d=1,2\dots,D-1.\tag{16d}$$

In both Scenarios 1 and 2, the optimization problem (1) is convex. According to the first-order optimality conditions, solving (1) is equivalent to finding $([x_d^\star]_{d\in\mathbb{N}_D},[y_d^\star]_{d\in\mathbb{N}_D})$ satisfying (16a)-(16d). Let $\mathcal{O}^\star$ be the set of all solutions of the KKT condition (16a)-(16d). The following theorem presents convergence to an optimal solution of the proposed hADMM in Algorithm 1 under Scenario 1.

Theorem 1. *Under Scenario 1, the sequence* $\{[x_d^k]_{d\in\mathbb{N}_D}, [y_d^k]_{d\in\mathbb{N}_{D-1}}\}_k$ *obtained by Algorithm 1 converges to some* $([x_d^\star]_{d\in\mathbb{N}_D}, [y_d^\star]_{d\in\mathbb{N}_{D-1}})$ *in* $\mathcal{O}^\star$.

Proof. See Appendix B. □

Similarly, the following theorem establishes the convergence of Algorithm 1 under Scenario 2, which assumes strong convexity of all local cost functions.

Theorem 2. *Under Scenario 2, the sequence* $\{[x_d^k]_{d\in\mathbb{N}_D}, [y_d^k]_{d\in\mathbb{N}_{D-1}}\}_k$ *obtained by Algorithm 1 converges to some* $([x_d^\star]_{d\in\mathbb{N}_D}, [y_d^\star]_{d\in\mathbb{N}_{D-1}})$ *in* $\mathcal{O}^\star$.

Proof. See Appendix C. □

B. Convergence rate of hADMM

To analyze the convergence rate, we consider the following scenario, which is similar to but stricter than Scenario 2 by assuming that all cost function gradients are Lipschitz continuous.

Scenario 3. *Function* f_1 *is differentiable and convex. Functions* $f_2, \dots f_D$ *are strongly convex with moduli* $\alpha_d > 0$*, and their gradient* ∇f_d *is Lipschitz continuous with constant* l_d*, that is, for all* $z, z' \in \mathrm{Dom}(f_d)$*,*

$$\|\nabla f_d(z) - \nabla f_d(z')\|_2 \le l_d \|z - z'\|_2. \tag{17}$$

Under Scenario 3, the hADMM has a linear convergence rate, as presented in the following theorem.

Theorem 3 (Convergence rate)**.** *Assume Scenario 3 and choose* $t \in (0,1)$ *and* $\sigma < \bar{\sigma} = \min_{d\in\mathbb{N}_{D-1}} \left\{\frac{2t\alpha_{d+1}}{\|B_d\|_2^2}\right\}$*. Let* $\gamma^\star$ *be the optimal solution of the following semidefinite programming (SDP) problem:*

$$\gamma^\star = \max_{\gamma>0} \gamma \tag{18a}$$

$$\text{s.t. } \Xi - \gamma\Theta \succeq 0, \tag{18b}$$

where $\Theta = \mathrm{diag}\left(\frac{1}{\rho}I_s, (\rho+\sigma)I_s, 0_s, 0_s\right)$,

$$\Xi = \left[\begin{array}{c|c:c:c} \Xi^{(1,1)} & 0 & 0 & (*) \\ \hline 0 & \Xi^{(2,2)} & 0 & (*) \\ 0 & 0 & \Xi^{(3,3)} & (*) \\ \Xi^{(4,1)} & \Xi^{(4,2)} & \Xi^{(4,3)} & \Xi^{(4,4)} \end{array}\right],$$

and $\Xi^{(1,1)}, \Xi^{(2,2)}, \Xi^{(3,3)}, \Xi^{(4,4)}, \Xi^{(4,1)}, \Xi^{(4,2)}, \Xi^{(4,3)}$ *are defined in* (19a) *to* (19e).

$$\Xi^{(1,1)} = 2(1-t)\,\mathrm{tridiag}\Bigg(\left\{\frac{1}{l_{i+1}}B_iA_{i+1}^\top\right\}_{i=1,\dots,D-2}, \left\{\frac{1}{l_2}B_1B_1^\top, \left\{\frac{1}{l_i}A_iA_i^\top + \frac{1}{l_{i+1}}B_iB_i^\top\right\}_{i=2,\dots,D-1}\right\}, \left\{\frac{1}{l_{i+1}}A_{i+1}^\top B_i\right\}_{i=1,\dots,D-2}\Bigg) \tag{19a}$$

$$\Xi^{(2,2)} = \left[\frac{2t\alpha_{d+1}}{\|B_d\|_2^2}I_{s_d}\right]_{d\in\mathbb{N}_{D-1}}^{\mathrm{diag}}, \quad \Xi^{(3,3)} = \rho I_s, \tag{19b}$$

$$\Xi^{(4,1)} = 2(1-t)\,\mathrm{tridiag}\Bigg(\left\{0\right\}_{i=1,\dots,D-2}, \left\{0, \left\{-\frac{1}{l_{i+1}}A_{i+1}B_i^\top\right\}_{i=1,\dots,D-2}\right\}, \left\{-\frac{1}{l_{i+1}}A_{i+1}B_i^\top\right\}_{i=1,\dots,D-2}\Bigg), \tag{19c}$$

$$\Xi^{(4,2)} = -\frac{\sigma}{\rho}I_s, \quad \Xi^{(4,3)} = -I_s, \tag{19d}$$

$$\Xi^{(4,4)} = \left(\frac{1}{\rho} + \frac{\sigma}{\rho^2}\right)I_s + \begin{bmatrix} 0 & \\ & \left[\frac{2(1-t)}{l_d}A_dA_d^\top\right]_{d=2,\dots,D-1}^{\mathrm{diag}} \end{bmatrix}. \tag{19e}$$

Then, the convergence rate of Algorithm 1 for the optimization problem (1) *satisfies*

$$(1+\gamma^\star)\hat{V}^{k+1} \le \hat{V}^k, \tag{20}$$

where $\hat{V}^k = \sum_{d=1}^{D-1} \frac{1}{\rho}\|y_d^k - y_d^\star\|_2^2 + (\rho+\sigma)\|B_d(x_{d+1}^k - x_{d+1}^\star)\|_2^2$ *represents the distance to the optimal solution.*

Proof. See Appendix D. □

The feasibility of (18) is crucial in Theorem 3. The following proposition provides a sufficient condition for its feasibility.

Proposition 1. *Suppose that the matrix* B_d *has full row rank for all* $d \in \mathbb{N}_{D-1}$*. Then, the SDP* (18) *is feasible.*

Proof. See Appendix E. □

Remark 1. *In Theorem 3, the SDP* (18) *maximizes the lower bound of the convergence rate* $\gamma^\star$ *with respect to the network structure, constraints, and penalty parameter* ρ*. This is essential for designing a tree network that maximizes the convergence rate. Furthermore, the choice of* (t, σ) *will affect* $\gamma^\star(t,\sigma)$*. To find the largest* $\gamma^\star$*, we can consider the bilevel optimization* $\max \gamma^\star(t,\sigma)$ s.t. $t \in (0,1), \sigma \in (0, \bar{\sigma})$.

Remark 2. *If* $D = 2$*, the network* $\mathcal{G}$ *has a star topology and the condition in Proposition 1 reduces to the requirement that* B_1 *has full row rank, which aligns with the conditions in [28, Assumption 4] and [29, Scenario 2].*

V. Numerical Examples

This section presents three numerical examples to validate the proposed hADMM method in Algorithm 1 and convergence results in Section IV. To compare our approach with existing methods, we consider two baselines:

- Baseline 1 is the nADMM described in Section II-C and Algorithm 2, which has been employed in [16], [17], [30].
- Baseline 2 is the fADMM outlined in Section II-D, which flattens the tree network for direct communication (via relays) between the root node and all the other nodes to apply the standard ADMM.

This paper considers the following three metrics to evaluate and compare the performance of the proposed hADMM with the baselines:

- *Maximum number of iterations in a node (`MaxNodeIters`):* This metric evaluates the convergence speed and the computational load of each method. In both hADMM and fADMM, all nodes perform the same number of iterations. In contrast, nADMM exhibits varying iteration counts across nodes due to the recursive structure described in Algorithm 2.
- *Maximum number of scalar values sent by a node (`MaxNodeVals`):* This metric quantifies the maximum total communication load experienced by a node, which may become a bottleneck due to excessive data transmission during the optimization process.
- *Total number of scalar values sent in the network (`TotalVals`):* This metric measures the overall communication cost across the network when solving the optimization problem (1). It reflects the average communication load and is approximately proportional to the total number of bits transmitted.

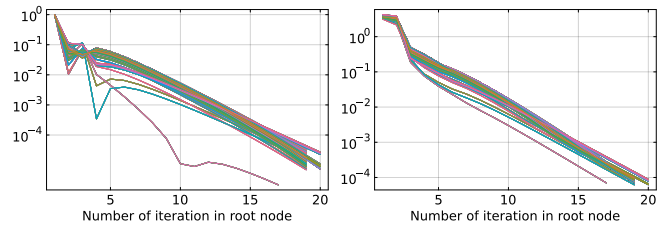
A. Disjoint problems over randomized tree networks

This experiment aims to thoroughly validate Algorithm 1 under various randomly generated tree-network topologies. Specifically, we used the proposed hADMM (Algorithm 1), nADMM (Algorithm 2), and fADMM for solving the disjoint problem (2) with three network configurations as $(D, N) = (3, 10), (3, 20)$ and $(5, 20)$. In each configuration, the algorithms are tested over 1000 randomized tree topologies to demonstrate the convergence behavior predicted by Theorem 1 and to compare the performance of these methods. In all cases, the matrices B_d have full column rank. Each local cost function is convex (but not strongly convex), and is defined as follows:

$$f_{d,i}(x_{d,i}) = \|x_{d,i} - u_{d,i}\|_2^4 + 2\|x_{d,i}\|_1,$$

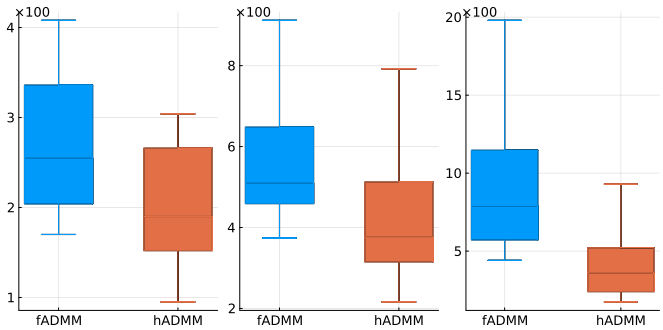
where $w > 0$ and $u_{d,i}$ is a real vector preassigned for each node. We can see that in this setup, Scenario 1 is satisfied. Each leaf node has a scalar local variable. The parameters for all algorithms are set as follows: $\rho = 100$ and $\epsilon = 10^{-4}$.

The convergence to an optimal solution of hADMM over 1000 tree networks with $(D = 3, N = 10)$ is shown in Fig. 2. From Fig. 2a, the normalized cost function error $\frac{J^k - J^\star}{J^\star}$ converges to zero as the iteration increases. These results validate the convergence proof established in Theorem 1 for non-smooth cost functions.



(a) Normalized cost function error (b) Primal residual

Fig. 2. Convergence of hADMM to an optimal solution under 1000 randomized tree networks with $(D = 3, N = 10)$.



(a) $D = 3, N = 10$ (b) $D = 3, N = 20$ (c) $D = 5, N = 20$

Fig. 3. Boxplots of `MaxNodeVals` of fADMM and hADMM under three network configurations, each is over 1000 randomized network topologies.

Table III summarizes the results of the performance metrics of the algorithms. As expected, nADMM consistently has the highest `MaxNodeIters` across all network configurations. This indicates that nADMM converges more slowly compared to fADMM and hADMM. Besides, nADMM also has the largest `MaxNodeVals` and `TotalVals`, which increase geometrically with the network depth, showing the highest and substantial communication costs among the methods.

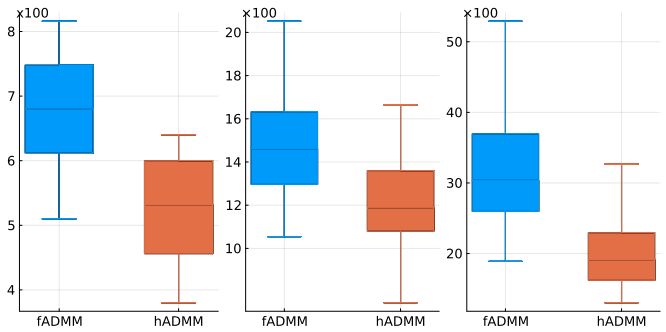
In Table III, we also observe that fADMM exhibits the smallest `MaxNodeIters`, as it is based on the star topology. However, its communication costs, measured by `MaxNodeVals` and `TotalVals`, are higher than those of hADMM. This is because nodes in lower layers (e.g., $d > 3$) incur more communication overhead when exchanging information with the root node, since all nodes along the paths between them must relay their messages. Statistical comparisons of communication costs between fADMM and hADMM are provided in the boxplots in Fig. 3 and Fig. 4. They clearly show that hADMM significantly reduces both `MaxNodeVals` and `TotalVals`, with the differences becoming increasingly pronounced as the network depth increases from $D = 3$ to $D = 5$.

B. Power sharing problem

Let us consider an electrical power distribution grid with a hierarchical structure described in the example in Section I. A power sharing problem is formulated for the grid, as presented in [17] and briefly summarized below. The utility company

TABLE III
PERFORMANCE COMPARISON FOR SOLVING THE DISJOINT PROBLEM (2) BETWEEN THE BASELINE METHODS (NADMM AND FADMM) AND OUR HADMM OVER 1000 RANDOMIZED TREE NETWORKS WITH DIFFERENT DEPTHS (D) AND NUMBERS OF NODES (N).

Performance metric	Network (depth and # nodes)	nADMM (min) mean (max)	fADMM (min) mean (max)	hADMM (min) mean (max)
Maximum Number of Iterations in a Node (`MaxNodeIters`)	D=3, N = 10	(56) 102 (111)	**(16) 17 (17)**	(17) 19 (20)
	D=3, N = 20	(51) 109 (148)	**(16) 17 (19)**	(16) 21 (27)
	D=5, N = 20	(483) 1463 (3735)	**(18) 21 (27)**	(24) 30 (40)
Maximum Number of scalar Values sent by a Node (`MaxNodeVals`)	D=3, N = 10	(353) 635 (1016)	(170) 255 (408)	**(95) 190 (304)**
	D=3, N = 20	(526) 1201 (2610)	(374) 510 (1026)	**(216) 376 (792)**
	D=5, N = 20	(902) 4740 (41168)	(420) 798 (2800)	**(174) 350 (1024)**
Total Number of scalar Values sent in network (`TotalVals`)	D=3, N = 10	(1103) 1951 (2792)	(510) 680 (816)	**(380) 532 (640)**
	D=3, N = 20	(1588) 3971 (7182)	(1020) 1462 (2052)	**(748) 1188 (1664)**
	D=5, N = 20	(7712) 28991 (126705)	(1760) 3040 (7840)	**(1218) 1860 (4096)**



(a) $D = 3, N = 10$ (b) $D = 3, N = 20$ (c) $D = 5, N = 20$

Fig. 4. Boxplots of `TotalVals` of fADMM and hADMM under three network configurations, each is over 1000 randomized network topologies.

coordinates with N_a aggregators, each managing multiple-device clusters to control power flows in the grid. Aggregators retain autonomy in managing and configuring these clusters. In real-world deployments, the power-flow control architectures and algorithms are secured between the aggregators and cluster layers and are not disclosed to the utility (root). Therefore, the utility can only communicate with the aggregators, not directly with the clusters. A power sharing problem optimally schedules the power allocations to all nodes in the network. Solving the sharing problem, formulated as a hierarchical optimization problem (1), enables the utility to coordinate power consumption across all aggregators to track a desired load profile while minimizing costs and satisfying all constraints at every level of the hierarchy.

For a time-dependent schedule with h_t horizon steps, denote $z_{ij} = [z_{ij}[t]]_{t=1\dots h_t} \in \mathbb{R}^{h_t}_{>0}$ as the vector of powers allocated to cluster j of aggregator i over h_t steps ahead. The agregator i is connected to $N_{c,i}$ clusters. We also denote $z_i[t] = \sum_{j=1}^{N_{c,i}} z_{ij}[t]$ as the power consumed by aggregator i at time step t, and $z_i = [z_i[t]]_{t=1\dots h_t}$. The nonnegative element-wise vector $\beta \in \mathbb{R}^{h_t}_{>0}$ stands for the amount of power obtained from the day-ahead wholesale market in order to satisfy the aggregator demand. For the total power to track the load signal β, costs are incurred at all three levels and our objective is to minimize the following total cost:

$$\min_{z_i \succeq 0} \sum_{i=1}^{N_a} f_i(z_i) + \lambda \Big\| \sum_{i=1}^{N_a} z_i - \beta \Big\|_2^2, \tag{21}$$

where $f_i(z_i)$ is the cost function of aggregator i and its clusters, and $\lambda \| \sum_{i=1}^{N_a} z_i - \beta \|_2^2$ is the utility-level penalty for deviation from the aggregate load signal with penalty factor $\lambda > 0$. Each aggregator aims to minimize both the penalty for drawing power beyond a predefined limit and the customer discomfort. Thus, the cost function f_i is given by:

$$f_i(z_i) = \sum_{j=1}^{N_{c,i}} f_{ij}(z_{ij}) + \eta \sum_{t=1}^{h_t} \max\left(\sum_{j=1}^{N_{c,i}} z_{ij}[t] - \tau_i, 0 \right),$$

where $f_{ij}(z_{ij}) = \sum_{t=1}^{h_t} c_{ij}(z_{ij}[t])$ with

$$c_{ij}(z_{ij}[t]) = \begin{cases} \frac{1}{z_{ij}[t]} - \frac{1}{a_{ij}}, & (0 < z_{ij}[t] < a_{ij}), \\ 0, & (z_{ij}[t] \geq a_{ij}), \end{cases}$$

standing for the customer discomfort in cluster j, τ_i is the upper limit of power consumption agreed upon between aggregator i and the utility, and η is the associated per-unit cost of exceeding this limit. In real-world applications, the optimal $z^\star_{ij}$ after solving (21) is sent to the corresponding cluster j of aggregator i. Then, controllers at the cluster level will use $z^\star_{ij}$ as a reference to regularize the power flow in the cluster.

To apply the hADMM to (21), we consider the power network as a tree network where the utility, aggregators, and clusters are the root, parent nodes, and leaf nodes, respectively. Let us define the problem variables and functions:

- At the root node: $x_{1,1} = [z_i]_{i=1\dots N_a}$ and $f_{1,1}(x_{1,1}) = \lambda \| \sum_{i=1}^{N_a} z_i - \beta \|_2^2$.
- At the aggregator nodes: $x_{2,i} = [z_{ij}]_{j=1\dots N_{c,i}}$ and $f_{2,i}(x_{2,i}) = \eta \sum_{t=1}^{h_t} \max\left(\sum_{j=1}^{N_{c,i}} z_{ij}[t] - \tau_i, 0 \right)$.
- At the cluster nodes: $x_{3,ij} = z_{ij}$ and $f_{3,ij} = f_{ij}$.

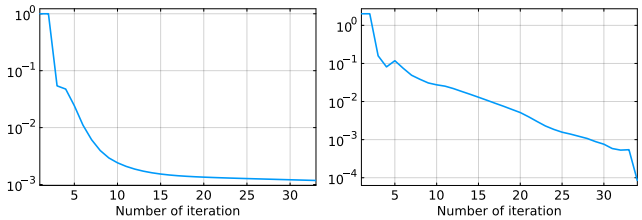
The sharing problem (21) is now written in the form of (1).

For our numerical experiment, the power network has three aggregators and each aggregator is connected to two clusters. The step size is 15 minutes and the planning period horizon is two hours ($h_t = 8$). The other parameters are $\beta = [28, 16, 20, 22, 18, 24, 15, 18]$, $\tau = [9, 8, 11]$, $\eta = 5$, $\lambda = 1$, the discomfort thresholds for the clusters in the three aggregators are $[a_{ij}] = [[2.0, 5.0], [3.0, 2.0], [3.0, 5.0]]$. For the hADMM, the penalty parameter is $\rho = 5$ and the stopping tolerance is $\varepsilon = 10^{-4}$.

Note that in this example, the cost functions are non-smooth and not strongly convex. The convergence of the hADMM for the power sharing experiment is shown in Fig. 5. Evidently,

TABLE IV
PERFORMANCE COMPARISON BETWEEN HADMM AND NADMM [17].

Performance metric	nADMM	hADMM
`MaxNodeIters`	86	34
`MaxNodeVals`	3584	1632
`TotalVals`	9808	6528



(a) Normalized cost functions error (b) Primal residual

Fig. 5. Convergence of the hADMM for the power sharing problem.

the normalized cost functions error and the primal residual converge as the number of iterations increases, validating Theorem 1. Furthermore, the hADMM has a significantly faster convergence rate and lower communication cost compared to the nADMM as shown in Table IV.

C. Convergence rate of hADMM for different topologies

This example aims to validate the convergence rate of the hADMM for different topologies. Two topologies are considered with the same numbers of levels and nodes as shown in Fig. 6. We apply hADMM to solve the following optimal consensus problem:

$$\text{minimize} \sum_{d=1}^{D} \sum_{i \in \mathcal{P}_d \cup \mathcal{L}_d} f_{d,i}(x_{d,i}) \tag{22a}$$

$$\text{s.t.}\, x_{d,i} = x_{d+1,j},\ \forall d \in \mathbb{N}_{D-1}, i \in \mathcal{P}_d, j \in \mathcal{C}_{d,i}, \tag{22b}$$

where $f_{d,i}$ is the elastic net regularization function: $f_{d,i}(x_{d,i}) = \frac{1}{2}||x_{d,i}-u||_2^2+\alpha_{d,i}||x_{d,i}||_1$. The cost functions are strongly convex with moduli $\alpha_d > 0$ and their gradients are Lipschitz continuous with Lipschitz constant $l_d = \alpha_d$. The cost functions (22a) are identical over the two network topologies, with $u = 2$ and $\alpha_{d,i} = 2$. The hADMM parameters are the same between the topologies: $\rho = 1$ and $\varepsilon = 10^{-4}$. The same initialization of the variables is used across the topologies. The theoretical convergence rates given by Theorem 3 are $\gamma^\star = 0.1144$ for Topology 1 and $\gamma^\star = 0.0208$ for Topology 2, where the largest $\gamma^\star$ is determined by bilevel optimization as discussed in Remark 1. As shown in Fig. 7, which plots $\hat{V}^k$ over the iterations, Topology 1 with higher $\gamma^\star$ converges over twice as fast as does Topology 2 with lower $\gamma^\star$ (22 iterations versus 55 iterations). Fig. 7 also shows that the convergence of $\hat{V}^k$ in both topologies is consistent with the theoretical convergence rates from Theorem 3.

VI. CONCLUSION

This paper addresses a general hierarchical optimization problem over tree-structured networks. The key technique lies in the reformulation of the augmented Lagrangian to explicitly exploit the hierarchical network structure, which enables the proposed hierarchical ADMM (hADMM) algorithm to solve the optimization problem in a fully distributed manner and at substantially reduced communication costs. The convergence guarantees and convergence rates of the hADMM are rigorously established under weak convexity assumptions. Through three numerical examples, the theoretical results are validated and the effectiveness of the proposed hADMM algorithm is demonstrated. Compared with existing approaches, such as the nested ADMM (nADMM) and the flattened ADMM (fADMM), hADMM achieves significantly lower communication costs at a convergence rate that is substantially faster than the nADMM and comparable to the fADMM, while removing the need for data-relaying mechanisms required by the fADMM. Future work will focus on integrating machine-learning-guided communication strategies to further improve the efficiency of the hADMM framework.

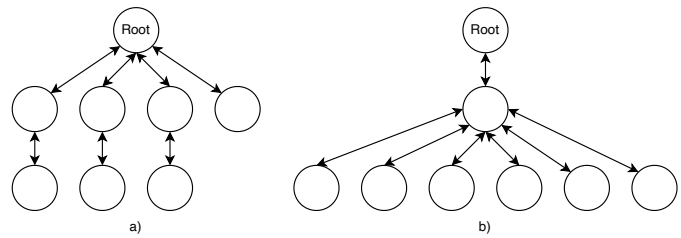


Fig. 6. Two topologies with $D = 3, N = 8$: a) Topology 1 has balance between levels 2 & 3, and b) Topology 2 has most nodes in the lowest level.

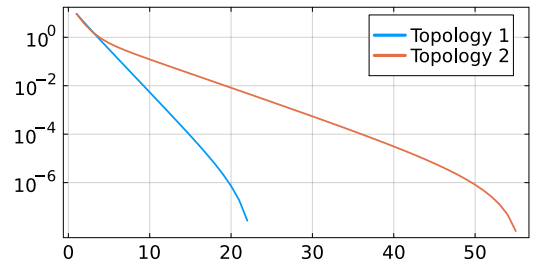


Fig. 7. Convergence of $\hat{V}^k$ by hADMM for the two topologies in Fig. 6.

APPENDIX A
THE GENERAL NADMM ALGORITHM

Let us define recursively the aggregate variable vector $X_{d,i} := \left[x_{d,i}^\top,\ [X_{d+1,j}]_{j\in\mathcal{C}_{d,i}}^\top\right]^\top$, that concatenates all the node variables in the subtree of node (d,i). For a leaf node (d,i), $X_{d,i} := x_{d,i}$. Similarly, define recursively the aggregate objective function of the subtree of node (d,i) as $F_{d,i}(X_{d,i}) = f_{d,i}(x_{d,i}) + \sum_{j\in\mathcal{C}_{d,i}} F_{d+1,j}(X_{d+1,j})$, where $F_{d,i}(X_{d,i}) = f_{d,i}(x_{d,i})$ for a leaf node. Finally, all the constraints (1b) in the subtree of a parent node (d,i) can be aggregated as $\mathcal{T}_{d,i}X_{d,i} = \tau_{d,i}$, which can be split recursively as $A_{d,ij}x_{d,i} + B_{d,ij}x_{d+1,j} = c_{d,ij}$ and $\mathcal{T}_{d+1,j}X_{d+1,j} = \tau_{d+1,j}$ for all $j \in \mathcal{C}_{d,i}$, where

$$\mathcal{T}_{d,i} = \begin{bmatrix} [A_{d,ij}]_{j\in\mathcal{C}_{d,i}} & [B_{d,ij}\ 0]_{j\in\mathcal{C}_{d,i}}^{\text{diag}} \\ 0 & [\mathcal{T}_{d+1,j}]_{j\in\mathcal{C}_{d,i}} \end{bmatrix},\ \tau_{d,i} = \begin{bmatrix} [c_{d,ij}]_{j\in\mathcal{C}_{d,i}} \\ [\tau_{d+1,j}]_{j\in\mathcal{C}_{d,i}} \end{bmatrix}.$$

Then, (1) is reformulated as

$$\underset{X_{1,1}}{\text{minimize}}\ F_{1,1}(X_{1,1}) = f_{1,1}(x_{1,1}) + \sum_{j\in\mathcal{C}_{1,1}} F_{2,j}(X_{2,j}), \quad (23a)$$

$$\text{s.t. } A_{1,1j}x_{1,1} + B_{1,1j}x_{2,j} = c_{1,1j}, \quad \forall j \in \mathcal{C}_{1,1}, \quad (23b)$$

$$\mathcal{T}_{2,j}X_{2,j} = \tau_{2,j}, \quad \forall j \in \mathcal{C}_{1,1}. \quad (23c)$$

Using the standard ADMM, (23) is solved by the iterations

$$x_{1,1}^{k+1} = \arg\min_{x_{1,1}} f_{1,1}(x_{1,1}) + \frac{\rho}{2}\|A_{1,1}x_{1,1} - v_{1,1}^k\|_2^2, \quad (24a)$$

$$X_{2,j}^{k+1} = \arg\min_{X_{2,j}} F_{2,j}(X_{2,j}) + \frac{\rho}{2}\|B_{1,1j}x_{2,j} - q_{1,1j}^{k+1}\|_2^2, \quad (24b)$$

$$\text{s.t. } \mathcal{T}_{2,j}X_{2,j} = \tau_{2,j},$$

$$y_{1,1j}^{k+1} = y_{1,1j}^k + \rho(A_{1,1j}x_{1,1}^{k+1} + B_{1,1j}x_{2,j}^{k+1} - c_{1,1j}), \quad (24c)$$

where $v_{1,1}^k = -[B_{1,i}]_{j\in\mathcal{C}_{1,i}}^{\text{diag}}[x_{2,j}^k]_{j\in\mathcal{C}_{1,1}} + c_{1,1} - \rho^{-1}y_{1,1}^k$ and $q_{1,1j}^{k+1} = -A_{1,1j}x_{1,i}^{k+1} + c_{1,1j} - \rho^{-1}y_{1,1j}^k$.

For each node $(2,j)$ at level $d=2$, let us define

$$\hat{f}_{2,j}(x_{2,j};q) = f_{2,j}(x_{2,j}) + \frac{\rho}{2}\|B_{1,1j}x_{2,j} - q\|_2^2.$$

If $(2,j)$ is a leaf node, (24b) becomes the proximal operator $\arg\min_{x_{2,j}} \hat{f}_{2,j}(x_{2,j}; q_{1,1j}^{k+1})$. If $(2,j)$ is a parent node, (24b) is equivalent to

$$\underset{X_{2,j}}{\text{minimize}}\ \hat{f}_{2,j}(x_{2,j}; q_{1,1j}^{k+1}) + \textstyle\sum_{l\in C_{2,j}} F_{3,l}(X_{3,l}), \quad (25a)$$

$$\text{s.t. } A_{2,jl}x_{2,j} + B_{2,jl}x_{3,l} = c_{2,jl}, \quad \forall l \in \mathcal{C}_{2,j}, \quad (25b)$$

$$\mathcal{T}_{3,l}X_{3,l} = \tau_{3,l}, \forall l \in \mathcal{C}_{2,j}. \quad (25c)$$

Observe that (25) has the same form as (23), hence it can be solved using the same algorithm in a recursive manner. This recursive, or nested, nADMM algorithm is formalized in Algorithm 2, where for $d = 2, \ldots, D-1$ and $j \in \mathcal{C}_{d,i}$,

$$v_{d,i}^k = -[B_{d,i}]_{j\in\mathcal{C}_{d,i}}^{\text{diag}}[x_{d+1,j}^k]_{j\in\mathcal{C}_{d,i}} + c_{d,i} - \rho^{-1}y_{d,i}^k, \quad (26a)$$

$$q_{d,ij}^{k+1} = -A_{d,ij}x_{d,i}^{k+1} + c_{d,ij} - \rho^{-1}y_{d,ij}^k, \quad (26b)$$

$$\hat{f}_{d+1,j}(x_{d+1,j};q) = f_{d+1,j}(x_{d+1,j}) + \frac{\rho}{2}\|B_{d,ij}x_{d+1,j} - q\|_2^2. \quad (26c)$$

Algorithm 2 Nested ADMM (nADMM) algorithm for hierarchical optimization problem (1)

function NADMM($(d,i), q$)
  if (d,i) is leaf node **then**
    return $\arg\min_x \hat{f}_{d,i}(x;q)$ ▷ with $\hat{f}_{d,i}$ in (26c)
  end if
  repeat with counter k starting from 1
    Compute $v_{d,i}^k$ in (26a)
    if (d,i) is root node **then**
      $x_{d,i}^{k+1} = \arg\min_x f_{d,i}(x) + \frac{\rho}{2}\|A_{d,i}x - v_{d,i}^k\|_2^2$
    else
      $x_{d,i}^{k+1} = \arg\min_x \hat{f}_{d,i}(x_{d,i};q) + \frac{\rho}{2}\|A_{d,i}x - v_{d,i}^k\|_2^2$
    end if
    for $j \in \mathcal{C}_{d,i}$ **do** ▷ In parallel for all j
      Compute $q_{d,ij}^{k+1}$ in (26b)
      $x_{d+1,j}^{k+1} =$ NADMM($(d+1,j)$, $q_{d,ij}^{k+1}$)
    end for
    $y_{d,ij}^{k+1} = y_{d,ij}^k + \rho(A_{d,ij}x_{d,i}^{k+1} + B_{d,ij}x_{d+1,j}^{k+1} - c_{d,ij})$
  until convergence
  return $x_{d,i}^{k+1}$
end function
NADMM($(1,1)$, []) ▷ Solve problem (1) at root node

Appendix B
Proof of Theorem 1

Lemma 1 ([31]). *For any convex function f, the following inequality holds $\forall z, z' \in \text{Dom}(f)$, $(\partial f(z) - \partial f(z'))^\top (z - z') \geq 0$.*

To simplify the presentation, let us denote:

$$r_d^{k+1} = A_d x_d^{k+1} + B_d x_{d+1}^{k+1} - c_d, \quad d = 1, \ldots, D-1, \quad (27)$$

$$J^k = \sum_{i=1}^{D} f_d(x_d^k), \quad J^\star = \sum_{i=1}^{D} f_d(x_d^\star), \quad (28)$$

$$V_d^k = \frac{1}{\rho}\|y_d^k - y_d^\star\|_2^2 + \rho\|B_d(x_{d+1}^k - x_{d+1}^\star)\|_2^2 \geq 0, \quad (29)$$

$$V^k = \sum_{d=1}^{D-1} V_d^k \geq 0. \quad (30)$$

Here, $([x_d^\star]_{d\in\mathbb{N}_D}, [y_d^\star]_{d\in\mathbb{N}_{D-1}})$ is any KKT point satisfying (16a)-(16d). We will prove the theorem in two steps.

1) **Step 1:** Inspired by the convergence proof of ADMM in [18], [20], we first prove that

$$V^k \geq V^{k+1} + \rho\sum_{d=1}^{D-1}\|r_d^{k+1} - B_d(x_{d+1}^{k+1} - x_{d+1}^k)\|_2^2. \quad (31)$$

2) **Step 2:** We then show that $\lim_{k\to\infty} x_d^k = x_d^\dagger$ for $d \in \mathbb{N}_D$ and $\lim_{k\to\infty} y_d^k = y_d^\dagger$ for all $d \in \mathbb{N}_{D-1}$, where $([x_d^\dagger]_{d\in\mathbb{N}_D}, [y_d^\dagger]_{d\in\mathbb{N}_{D-1}}) \in \mathcal{O}^\star$ is an optimal solution.

We now detail these steps.

Step 1: After the forward pass (9), x_d^{k+1} satisfies the generalized first-order condition [32]:

$$0 \in \partial f_1(x_1^{k+1}) + A_1^\top(y_1^k + \rho(A_1x_1^{k+1} + B_1x_2^k - c_1)),$$
$$0 \in \partial f_D(x_D^{k+1}) + B_{D-1}^\top(y_{D-1}^k + \rho(A_{D-1}x_{D-1}^{k+1} + B_{D-1}x_D^{k+1} - c_{D-1})),$$
$$0 \in \partial f_d(x_d^{k+1}) + A_d^\top(y_d^k + \rho(A_dx_d^{k+1} + B_dx_{d+1}^k - c_d)) + B_{d-1}^\top(y_{d-1}^k + \rho(A_{d-1}x_{d-1}^{k+1} + B_{d-1}x_d^{k+1} - c_{d-1})),$$

for $1 < d < D$. Substituting (27) into these conditions and noting that $y_d^{k+1} = y_d^k + \rho r_d^{k+1}$ from the backward pass (10), we have

$$0 \in \partial f_1(x_1^{k+1}) + A_1^\top(y_1^{k+1} - \rho B_1(x_2^{k+1} - x_2^k)), \quad (32a)$$

$$0 \in \partial f_D(x_D^{k+1}) + B_{D-1}^\top y_{D-1}^{k+1}, \quad (32b)$$

$$0 \in \partial f_d(x_d^{k+1}) + A_d^\top(y_d^{k+1} - \rho B_d(x_{d+1}^{k+1} - x_{d+1}^k)) + B_{d-1}^\top y_{d-1}^{k+1}, \quad 1 < d < D. \quad (32c)$$

Because of the convexity of f_d, x_d^{k+1} minimizes the following functions

$$f_1(x)+(y_1^{k+1}-\rho B_1(x_2^{k+1}-x_2^k))^\top A_1 x,$$
$$f_D(x)+\left(y_{D-1}^{k+1}\right)^\top B_{D-1}x,$$
$$f_d(x)+\left(A_d^\top(y_d^{k+1}-\rho B_d(x_{d+1}^{k+1}-x_{d+1}^k))+B_{d-1}^\top y_{d-1}^{k+1}\right)^\top x$$

for $1<d<D$. Thus, the following inequalities hold

$$f_d(x_d^{k+1})+\Theta_d^{k+1}x_d^{k+1}\le f_d(x_d^\star)+\Theta_d^{k+1}x_d^\star,$$

where $\Theta_1^{k+1} = \left(y_1^{k+1}-\rho B_1(x_2^{k+1}-x_2^k)\right)^\top A_1, \Theta_D^{k+1} = \left(y_{D-1}^{k+1}\right)^\top B_{D-1}$, and $\Theta_d^{k+1}=(y_d^{k+1}-\rho B_d(x_{d+1}^{k+1}-x_{d+1}^k))^\top A_d+(y_{d-1}^{k+1})^\top B_{d-1}$ for $1<d<D$. Summing the above inequalities for all $d=1,\dots,D$ and using (28), we have

$$J^{k+1}\le J^\star+g^{k+1}, \tag{33}$$

where

$$\begin{aligned} g^{k+1} &= \sum_{d=1}^{D}(\Theta_d^{k+1})^\top(x_d^\star-x_d^{k+1})\\ &= (y_{D-1}^{k+1})^\top B_{D-1}(x_D^\star-x_D^{k+1})+(y_1^{k+1})^\top A_1(x_1^\star-x_1^{k+1})\\ &\quad+\sum_{d=1}^{D-1}\rho(x_{d+1}^{k+1}-x_{d+1}^k)^\top B_d^\top A_d(x_d^{k+1}-x_d^\star)\\ &\quad+\sum_{d=2}^{D-1}(A_d^\top y_d^{k+1}+B_{d-1}^\top y_{d-1}^{k+1})^\top(x_d^\star-x_d^{k+1}). \end{aligned} \tag{34}$$

By the KKT condition (16d) and by (27), we can derive

$$A_d(x_d^\star-x_d^{k+1})+B_d(x_{d+1}^\star-x_{d+1}^{k+1})=-r_d^{k+1}. \tag{35}$$

Left-multiplying both sides by $(y_d^{k+1})^\top$ then summing the equations for all d from 1 to $D-2$, noting that $(y_{D-1}^{k+1})^\top B_{D-1}(x_D^\star-x_D^{k+1}) = (y_{D-1}^{k+1})^\top(-r_{D-1}^{k+1}-A_{D-1}(x_{D-1}^\star-x_{D-1}^{k+1}))$ from (35), we can rewrite (34) as

$$\begin{aligned} g^{k+1} = \sum_{d=1}^{D-1}\Big(&\rho(x_{d+1}^{k+1}-x_{d+1}^k)^\top B_d^\top A_d(x_d^{k+1}-x_d^\star)\\ &-(y_d^{k+1})^\top r_d^{k+1}\Big). \end{aligned} \tag{36}$$

Since $(x^\star,y^\star)$ is a saddle point for the augmented Lagrangian L at any ρ (see [31, Sec 5.4]), from (7) and (27), we have

$$\begin{aligned} J^\star = L(x^\star,y^\star)\Big|_{\rho=0} &\le L(x^{k+1},y^\star)\Big|_{\rho=0}\\ &= J^{k+1}+\sum_{d=1}^{D-1}\left(y_d^\star\right)^\top r_d^{k+1}. \end{aligned}$$

Then, combining with (33), we obtain

$$-\sum_{i=0}^{D-1}\left(y_d^\star\right)^\top r_d^{k+1}\le J^{k+1}-J^\star\le g^{k+1}. \tag{37}$$

This implies $0\le 2\left(g^{k+1}+\sum_{d=1}^{D-1}y_d^{\star\top}r_d^{k+1}\right)$, which from (36) gives

$$\begin{aligned} 0\le\sum_{d=1}^{D-1}\Big(&2\rho(x_{d+1}^{k+1}-x_{d+1}^k)^\top B_d^\top A_d(x_d^{k+1}-x_d^\star)\\ &-2(y_d^{k+1}-y_d^\star)^\top r_d^{k+1}\Big). \end{aligned} \tag{38}$$

We will rewrite the right hand side (RHS) of (38) in terms of V^k and V^{k+1}. First, using (35), the first term in the summand can be expressed as

$$\begin{aligned} &2\rho(x_{d+1}^{k+1}-x_{d+1}^k)^\top B_d^\top A_d(x_d^{k+1}-x_d^\star)\\ &=2\rho(x_{d+1}^{k+1}-x_{d+1}^k)^\top B_d^\top(r_d^{k+1}-B_d(x_{d+1}^{k+1}-x_{d+1}^\star))\\ &=2\rho(x_{d+1}^{k+1}-x_{d+1}^k)^\top B_d^\top r_d^{k+1}\\ &\quad-2\rho(x_{d+1}^{k+1}-x_{d+1}^k)^\top B_d^\top B_d(x_{d+1}^{k+1}-x_{d+1}^\star)\\ &=\rho\Big(2(x_{d+1}^{k+1}-x_{d+1}^k)^\top B_d^\top r_d^{k+1}-\|B_d(x_{d+1}^{k+1}-x_{d+1}^k)\|_2^2\\ &\quad+\|B_d(x_{d+1}^k-x_{d+1}^\star)\|_2^2-\|B_d(x_{d+1}^{k+1}-x_{d+1}^\star)\|_2^2\Big)\\ &=-\rho\|r_d^{k+1}-B_d(x_{d+1}^{k+1}-x_{d+1}^k)\|_2^2+\rho\|r_d^{k+1}\|_2^2\\ &\quad+\rho\|B_d(x_{d+1}^k-x_{d+1}^\star)\|_2^2-\rho\|B_d(x_{d+1}^{k+1}-x_{d+1}^\star)\|_2^2. \end{aligned}$$

Using $y_d^{k+1}=y_d^k+\rho r_d^{k+1}$, the second term in the summand of (38) can be rewritten as

$$\begin{aligned} &2(y_d^{k+1}-y_d^\star)^\top r_d^{k+1}\\ &=2(y_d^k-y_d^\star)^\top r_d^{k+1}+2\rho\|r_d^{k+1}\|_2^2\\ &=\frac{2}{\rho}(y_d^k-y_d^\star)^\top(y_d^{k+1}-y_d^k)+2\rho\|r_d^{k+1}\|_2^2\\ &=-\frac{2}{\rho}(y_d^k-y_d^\star)^\top(y_d^\star-y_d^{k+1})-\frac{2}{\rho}\|y_d^k-y_d^\star\|_2^2+2\rho\|r_d^{k+1}\|_2^2\\ &=\frac{1}{\rho}\|y_d^{k+1}-y_d^\star\|_2^2-\frac{1}{\rho}\|y_d^k-y_d^\star\|_2^2+\rho\|r_d^{k+1}\|_2^2. \end{aligned}$$

Using these expressions and (29), the summand in (38) is rewritten as

$$\begin{aligned} &2\rho(x_{d+1}^{k+1}-x_{d+1}^k)^\top B_d^\top A_d(x_d^{k+1}-x_d^\star)-2(y_d^{k+1}-y_d^\star)^\top r_d^{k+1}\\ &\quad=V_d^k-V_d^{k+1}-\rho\|r_d^{k+1}-B_d(x_{d+1}^{k+1}-x_{d+1}^k)\|_2^2. \end{aligned} \tag{39}$$

Summing (39) for all d from 1 to $D-1$ and using (30) and (38), we have (31).

Step 2: According to (32b), x_D^{k+1} and x_D^k are minimizers of $f_D(x)+(y_{D-1}^{k+1})^\top B_{D-1}x$ and $f_D(x)+(y_{D-1}^k)^\top B_{D-1}x$, respectively. Thus, we can derive $f_D(x_D^{k+1})-f_D(x_D^k)+(y_{D-1}^{k+1})^\top B_{D-1}(x_D^{k+1}-x_D^k)\le 0$, and $f_D(x_D^k)-f_D(x_D^{k+1})+(y_{D-1}^k)^\top B_{D-1}(x_D^k-x_D^{k+1})\le 0$. Adding the inequalities leads to $(r_{D-1}^{k+1})^\top B_{D-1}(x_D^{k+1}-x_D^k)\le 0$. This implies $\|r_{D-1}^{k+1}-B_{D-1}(x_D^{k+1}-x_D^k)\|_2^2\ge\|r_{D-1}^{k+1}\|_2^2+\|B_{D-1}(x_D^{k+1}-x_D^k)\|_2^2$. Then, from (31), we have

$$\begin{aligned} V^k\ge V^{k+1}+\rho\sum_{d=1}^{D-2}&\|r_d^{k+1}-B_d(x_{d+1}^{k+1}-x_{d+1}^k)\|_2^2\\ &+\rho\|r_{D-1}^{k+1}\|_2^2+\rho\|B_{D-1}(x_D^{k+1}-x_D^k)\|_2^2. \end{aligned}$$

Also from (31), V^k is a non-increasing non-negative sequence and hence converges to a constant. Therefore, $r_{D-1}^{k+1}\to 0$, $B_{D-1}(x_D^{k+1}-x_D^k)\to 0$, and $r_d^{k+1}-B_d(x_{d+1}^{k+1}-x_{d+1}^k)\to 0$ for $d=1,\dots,D-2$. Since B_{D-1} is a full-rank matrix in Scenario 1, $x_D^{k+1}-x_D^k\to 0$ or x_D^k converges to a constant vector $x_D^\dagger$. Moreover, because $r_{D-1}^{k+1}\to 0$, $y_{D-1}^{k+1}-y_{D-1}^k\to 0$ or y_{D-1}^{k+1} converges to a constant vector $y_{D-1}^\dagger$.

Similarly, (32c) holds for the previous iteration k (instead of $k+1$). Subtracting the two equations at $d = D-1$, we obtain

$$0 \in \partial f_{D-1}(x_{D-1}^{k+1}) - \partial f_{D-1}(x_{D-1}^{k}) + A_{D-1}^\top (y_{D-1}^{k+1} - y_{D-1}^{k}) + B_{D-2}(y_{D-2}^{k+1} - y_{D-2}^{k}) - \rho A_{D-1}^\top B_{D-1}(x_D^{k+1} - x_D^{k-1}).$$

Pre-multiplying the equation by $(x_{D-1}^{k+1} - x_{D-1}^{k})^\top$ then applying Lemma 1 yields

$$\begin{aligned} 0 \geq\ & (x_{D-1}^{k+1} - x_{D-1}^{k})^\top B_{D-2}(y_{D-2}^{k+1} - y_{D-2}^{k}) \\ & + (x_{D-1}^{k+1} - x_{D-1}^{k})^\top A_{D-1}^\top (y_{D-1}^{k+1} - y_{D-1}^{k}) \\ & - \rho (x_{D-1}^{k+1} - x_{D-1}^{k})^\top A_{D-1}^\top B_{D-1}(x_D^{k+1} - x_D^{k-1}). \end{aligned} \tag{40}$$

Taking the limit of the RHS of (40) as $k \to \infty$, the last two terms approach 0 because $x_D^k \to x_D^\dagger$ and $y_{D-1}^{k+1} \to y_{D-1}^\dagger$. Noting that $y_d^{k+1} - y_d^k = \rho r_d^{k+1}$, we then have

$$0 \geq \lim_{k\to\infty} \left((x_{D-1}^{k+1} - x_{D-1}^{k})^\top B_{D-2} r_{D-2}^{k+1} \right). \tag{41}$$

Recall that $\|r_d^{k+1} - B_d(x_{d+1}^{k+1} - x_{d+1}^k)\|_2^2 \to 0$. Using (41), it follows that $\|r_{D-2}^{k+1}\|_2 \to 0$ and $\|x_{D-1}^{k+1} - x_{D-1}^k\|_2 \to 0$. As a result, $y_{D-2}^k \to y_{D-2}^\dagger$ and $x_{D-1}^k \to x_{D-1}^\dagger$. Repeat the above argument for $d = D-2, \ldots, 1$, we can show that $y_d^k \to y_d^\dagger$ for all $d = 1, \ldots, D-1$ and $x_d^k \to x_d^\dagger$ for all $d = 1, \ldots, D$. It is noted that all $x_d^\dagger$ satisfies the constraints (6b). In addition, at the limit of (32a)-(32c) as $k \to \infty$, $x_d^\dagger$ and $y_d^\dagger$ satisfy the KKT conditions (16a)-(16d). Therefore, $[x_d^\dagger]_{d\in\mathbb{N}_D}$ and $[y_d^\dagger]_{d\in\mathbb{N}_{D-1}}$ is an optimal primal and a dual solution of the optimization problem (1).

Appendix C
Proof of Theorem 2

Using the same definitions in (27) to (30) in Appendix B, we will show that

$$V^k - V^{k+1} \geq \rho \sum_{d=1}^{D-1} \|r_d^{k+1} - B_d(x_{d+1}^{k+1} - x_{d+1}^k)\|_2^2 + 2\sum_{d=2}^{D} \alpha_d \|x_d^{k+1} - x_d^\star\|_2^2. \tag{42}$$

If (42) holds, by similar arguments as in Appendix B, $x_d^{k+1} - x_d^\star \to 0$ for $d = 1, \ldots, D$ and $r_d^{k+1} - B_d(x_{d+1}^{k+1} - x_{d+1}^k) \to 0$ for $d = 1, \ldots, D-1$ as $k \to \infty$, regardless of A_d and B_d. Consequently, $r_d^{k+1} \to 0$ or $y_d^{k+1} \to y_d^\star$ for all $d = 1, \ldots, D-1$.

To prove (42), let us consider the following inequality obtained by the strong convexity of f_d in Scenario 2:

$$\begin{aligned} J^{k+1} - J^\star &= \sum_{d=1}^{D} \left(f_d(x_d^{k+1}) - f_d(x_d^\star) \right) \\ &\geq \sum_{d=1}^{D} \partial f_d^\top(x_d^\star)(x_d^{k+1} - x_d^\star) + \sum_{d=2}^{D} \alpha_d \|x_d^{k+1} - x_d^\star\|_2^2. \end{aligned} \tag{43}$$

Recalling g^{k+1} defined in (34) then using (39), we have

$$g^{k+1} + \sum_{d=1}^{D-1} y_d^{\star\top} r_d^{k+1} = \frac{1}{2}\left(V^k - V^{k+1}\right) - \frac{\rho}{2} \sum_{d=1}^{D-1} \|r_d^{k+1} - B_d(x_{d+1}^{k+1} - x_{d+1}^k)\|_2^2. \tag{44}$$

On the other hand, according to (37), we have that

$$g^{k+1} \geq J^{k+1} - J^\star.$$

Applying (43) and (44) into this inequality leads to

$$\begin{aligned} \frac{1}{2}(V^k - V^{k+1}) \geq\ & \sum_{d=1}^{D-1} y_d^{\star\top} r_d^{k+1} + \sum_{d=1}^{D} \partial f_d^\top(x_d^\star)(x_d^{k+1} - x_d^\star) \\ & + \frac{\rho}{2} \sum_{d=1}^{D-1} \|r_d^{k+1} - B_d(x_{d+1}^{k+1} - x_{d+1}^k)\|_2^2 \\ & + \sum_{d=2}^{D} \alpha_d \|x_d^{k+1} - x_d^\star\|_2^2. \end{aligned} \tag{45}$$

From (16), the first two terms in the RHS of (45) reduce to

$$\begin{aligned} & \sum_{d=1}^{D-1} \left(y_d^{\star\top} r_d^{k+1} + \partial f_d^\top(x_d^\star)(x_d^{k+1} - x_d^\star) \right) + \partial f_D^\top(x_D^\star)(x_D^{k+1} - x_D^\star) \\ =\ & \sum_{d=1}^{D-1} \left(y_d^{\star\top} r_d^{k+1} - y_d^{\star\top} A_d (x_d^{k+1} - x_d^\star) \right) - \sum_{d=1}^{D-1} y_d^{\star\top} B_d^\top (x_{d+1}^{k+1} - x_{d+1}^\star) \\ =\ & \sum_{d=1}^{D-1} \left(y_d^{\star\top} (B_d x_{d+1}^{k+1} - c_d) + y_d^{\star\top} A_d x_d^\star \right) - \sum_{d=1}^{D-1} y_d^{\star\top} B_d^\top (x_{d+1}^{k+1} - x_{d+1}^\star) \\ =\ & \sum_{d=1}^{D-1} \left(y_d^{\star\top} B_d x_{d+1}^{k+1} - y_d^{\star\top} B_d x_{d+1}^\star \right) - \sum_{d=1}^{D-1} y_d^{\star\top} B_d^\top (x_{d+1}^{k+1} - x_{d+1}^\star) \\ =\ & 0. \end{aligned}$$

Therefore, from (45), (42) is proved.

Appendix D
Proof of Theorem 3

Lemma 2 ([28], [29])**.** *For any strongly convex function f with moduli $\alpha > 0$, the following inequality holds for all $z, z' \in \mathrm{Dom}(f)$, $\alpha\|z - z'\|_2^2 \leq (\nabla f(z) - \nabla f(z'))^\top (z - z')$.*

Lemma 3 ([28], [29])**.** *For any differentiable convex function f having Lipschitz continuous gradient with Lipschitz constant L, the following inequality holds for all $z, z' \in \mathrm{Dom}(f)$, $\|\nabla f(z) - \nabla f(z')\|_2^2 \leq L(z - z')^\top (\nabla f(z) - \nabla f(z'))$.*

To streamline the presentation, let us denote

$$\begin{aligned}
&\tilde{y}_d^k = y_d^k - y_d^\star,\ \tilde{x}_d^k = B_d(x_d^k - x_d^\star),\\
&\delta_d^{k+1} = \rho B_d(x_d^{k+1} - x_d^k),\\
&\boldsymbol{\eta}^k = \big[[\tilde{y}_d^k]_{d\in\mathbb{N}_{D-1}}, [\tilde{x}_{d+1}^k]_{d\in\mathbb{N}_{D-1}}, [r_d^k]_{d\in\mathbb{N}_{D-1}}^\top, [\delta_{d+1}^k]_{d\in\mathbb{N}_{D-1}}^\top\big]
\end{aligned}$$

where $([x_d^\star]_{d\in\mathbb{N}_D}, [y_d^\star]_{d\in\mathbb{N}_{D-1}})$ is a KKT solution of (16a)-(16d), the residual r_d^k is defined in (27), and $\boldsymbol{\eta}^k$ collects all the variables. With the two lemmas above, we will prove that

$$\begin{aligned}
&\hat{V}^k - \hat{V}^{k+1}\\
&\geq \frac{2(1-t)}{l_D}\|B_{D-1}^\top \tilde{y}_{D-1}^{k+1}\|_2^2 + \sum_{d=1}^{D-1}\left(\frac{2t\alpha_{d+1}}{\|B_d\|_2^2} - \sigma\right)\|\tilde{x}_{d+1}^{k+1}\|_2^2\\
&\quad + \sum_{d=2}^{D-1}\frac{2(1-t)}{l_d}\|A_d^\top \tilde{y}_d^{k+1} + B_{d-1}^\top \tilde{y}_{d-1}^{k+1} - A_d^\top \delta_{d+1}^{k+1}\|_2^2\\
&\quad + \rho\|r_d^{k+1} - \frac{1}{\rho}\delta_{d+1}^{k+1}\|_2^2 + \sigma\|\tilde{x}_{d+1}^{k+1} - \frac{1}{\rho}\delta_{d+1}^{k+1}\|_2^2\\
&= (\boldsymbol{\eta}^{k+1})^\top \Xi \boldsymbol{\eta}^{k+1}.
\end{aligned} \tag{46}$$

From the definitions of $\hat{V}^{k+1}$ and Θ in Theorem 3, we have $\hat{V}^{k+1} = (\boldsymbol{\eta}^{k+1})^\top \Theta \boldsymbol{\eta}^{k+1}$. If the inequality (46) holds, by adding $0 = \gamma^\star \hat{V}^{k+1} - \gamma^\star (\boldsymbol{\eta}^{k+1})^\top \Theta \boldsymbol{\eta}^{k+1}$ to the RHS of (46), the convergence rate $(1+\gamma^\star)\hat{V}^{k+1} \leq \hat{V}^k$ given by (18) is proved. We will now prove (46).

For brevity, denote $f_d^k = f_d(x_d^k)$, $f_d^\star = f_d(x_d^\star)$, $\nabla f_d^k = \nabla f_d(x_d^k)$ and $\nabla f_d^\star = \nabla f_d(x_d^\star)$. Under Scenario 3, due to the strong convexity of f_d and the Lipschitz continuity of ∇f_d, Lemmas 3 and 2 lead to, for all $1 \leq d \leq D$,

$$\|\nabla f_d^{k+1} - \nabla f_d^\star\|_2^2 \leq l_d (x_d^{k+1} - x_d^\star)^\top (\nabla f_d^{k+1} - \nabla f_d^\star), \tag{47}$$

and

$$\alpha_d \|x_d^{k+1} - x_d^\star\|_2^2 \leq (\nabla f_d^{k+1} - \nabla f_d^\star)^\top (x_d^{k+1} - x_d^\star). \tag{48}$$

From (16a)-(16d) and (32a)-(32c), we have

$$\nabla f_1^{k+1} - \nabla f_1^\star = -A_1^\top \tilde{y}_1^{k+1} + A_1^\top \delta_2^{k+1}, \tag{49}$$
$$\nabla f_D^{k+1} - \nabla f_D^\star = -B_{D-1}^\top \tilde{y}_{D-1}^{k+1}, \tag{50}$$
$$\nabla f_d^{k+1} - \nabla f_d^\star = -B_{d-1}^\top \tilde{y}_{d-1}^{k+1} - A_d^\top \tilde{y}_d^{k+1} + A_d^\top \delta_{d+1}^{k+1}, \tag{51}$$

for $0 \leq d \leq D-1$. Then we can rewrite (47) and (48) with the help of (49)-(51) as follows. At $d = D$,

$$0 \geq \frac{1}{l_D}\|B_{D-1}^\top \tilde{y}_{D-1}^{k+1}\|_2^2 + (\tilde{x}_D^{k+1})^\top \tilde{y}_{D-1}^{k+1}, \tag{52}$$

$$\begin{aligned}
0 &\geq \alpha_D \|x_D^{k+1} - x_D^\star\|_2^2 + (\tilde{x}_D^{k+1})^\top \tilde{y}_{D-1}^{k+1}\\
&\geq \frac{\alpha_D}{\|B_{D-1}\|_2^2}\|\tilde{x}_D^{k+1}\|_2^2 + (\tilde{x}_D^{k+1})^\top \tilde{y}_{D-1}^{k+1}.
\end{aligned} \tag{53}$$

At $d = 2, \ldots, D-1$, noting that $A_d(x_d^{k+1} - x_d^\star) = r_d^{k+1} - B_d(x_{d+1}^{k+1} - x_{d+1}^\star) = r_d^{k+1} - \tilde{x}_{d+1}^{k+1}$, we have

$$\begin{aligned}
0 &\geq \frac{1}{l_d}\|A_d^\top \tilde{y}_d^{k+1} + B_{d-1}^\top \tilde{y}_{d-1}^{k+1} - A_d^\top \delta_{d+1}^{k+1}\|_2^2\\
&\quad + (x_d^{k+1} - x_d^\star)^\top \Big(A_d^\top \tilde{y}_d^{k+1} + B_{d-1}^\top \tilde{y}_{d-1}^{k+1} - A_d^\top \delta_{d+1}^{k+1}\Big)\\
&= \frac{1}{l_d}\|A_d^\top \tilde{y}_d^{k+1} + B_{d-1}^\top \tilde{y}_{d-1}^{k+1} - A_d^\top \delta_{d+1}^{k+1}\|_2^2 + (\tilde{x}_d^{k+1})^\top \tilde{y}_{d-1}^{k+1}\\
&\quad + (r_d^{k+1} - \tilde{x}_{d+1}^{k+1})^\top \tilde{y}_d^{k+1} - (x_d^{k+1} - x_d^\star)^\top \delta_{d+1}^{k+1}\\
&\underset{(39)}{=} \frac{1}{l_d}\|A_d^\top \tilde{y}_d^{k+1} + B_{d-1}^\top \tilde{y}_{d-1}^{k+1} - A_d^\top \delta_{d+1}^{k+1}\|_2^2 + (\tilde{x}_d^{k+1})^\top \tilde{y}_{d-1}^{k+1}\\
&\quad - (\tilde{x}_{d+1}^{k+1})^\top \tilde{y}_d^{k+1} - \frac{1}{2}(V_d^k - V_d^{k+1} - \rho\|r_d^{k+1} - \frac{1}{\rho}\delta_{d+1}^{k+1}\|_2^2).
\end{aligned} \tag{54}$$

$$\begin{aligned}
0 &\geq \alpha_d \|x_d^{k+1} - x_d^\star\|_2^2\\
&\quad + (x_d^{k+1} - x_d^\star)^\top \Big(A_d^\top \tilde{y}_d^{k+1} + B_{d-1}^\top \tilde{y}_{d-1}^{k+1} - A_d^\top \delta_{d+1}^{k+1}\Big)\\
&\geq \frac{\alpha_d}{\|B_{d-1}\|_2^2}\|\tilde{x}_d^{k+1}\|_2^2 + (\tilde{x}_d^{k+1})^\top \tilde{y}_{d-1}^{k+1}\\
&\quad + (r_d^{k+1} - \tilde{x}_{d+1}^{k+1})^\top \tilde{y}_d^{k+1} - (x_d^{k+1} - x_d^\star)^\top \delta_{d+1}^{k+1}\\
&\underset{(39)}{=} \frac{\alpha_d}{\|B_{d-1}\|_2^2}\|\tilde{x}_d^{k+1}\|_2^2 + (\tilde{x}_d^{k+1})^\top \tilde{y}_{d-1}^{k+1} - (\tilde{x}_{d+1}^{k+1})^\top \tilde{y}_d^{k+1}\\
&\quad - \frac{1}{2}(V_d^k - V_d^{k+1} - \rho\|r_d^{k+1} - \frac{1}{\rho}\delta_{d+1}^{k+1}\|_2^2).
\end{aligned} \tag{55}$$

where V_d^k is defined in (29). Moreover, because of the convexity of f_1, it holds that

$$\begin{aligned}
0 &\geq (x_1^{k+1} - x_1^\star)^\top \Big(A_1^\top \tilde{y}_1^{k+1} - A_2^\top \delta_2^{k+1}\Big),\\
&= (r_1^{k+1} - \tilde{x}_2^{k+1})^\top \tilde{y}_1^{k+1} - (x_1^{k+1} - x_1^\star)^\top \delta_2^{k+1}\\
&\underset{(39)}{=} -(\tilde{x}_2^{k+1})^\top \tilde{y}_1^{k+1}\\
&\quad - \frac{1}{2}(V_1^k - V_1^{k+1} - \rho\|r_1^{k+1} - \frac{1}{\rho}\delta_2^{k+1}\|_2^2).
\end{aligned} \tag{56}$$

Multiplying (53) and (55) by $t \in (0,1)$, and multiplying (52) and (54) by $(1-t)$, then adding the results and (56) together, we obtain the following inequality

$$\begin{aligned}
&\sum_{d=1}^{D-1} V_d^k - V_d^{k+1}\\
&\geq \frac{2(1-t)}{l_D}\|B_{D-1}^\top \tilde{y}_{D-1}^{k+1}\|_2^2\\
&\quad + \sum_{d=2}^{D-1}\frac{2(1-t)}{l_d}\|A_d^\top \tilde{y}_d^{k+1} + B_{d-1}^\top \tilde{y}_{d-1}^{k+1} - A_d^\top \delta_{d+1}^{k+1}\|_2^2\\
&\quad + \sum_{d=1}^{D-1}\frac{2t\alpha_{d+1}}{\|B_d\|_2^2}\|\tilde{x}_{d+1}^{k+1}\|_2^2 + \rho\|r_d^{k+1} - \frac{1}{\rho}\delta_{d+1}^{k+1}\|_2^2.
\end{aligned}$$

Adding $\sigma\sum_{d=1}^{D-1}(\|\tilde{x}_{d+1}^k\|_2^2 - \|\tilde{x}_{d+1}^{k+1}\|_2^2)$ to both sides of the inequality and noting that $\|\tilde{x}_d^k\|_2^2 = \|\tilde{x}_d^{k+1} - \frac{1}{\rho}\delta_d^{k+1}\|_2^2$, we now obtain (46).

Appendix E
Proof of Proposition 1

Proposition 1 is proved by showing that $\Xi \succ 0$.

We will first show that $\Xi \succeq 0$. From the convergence proof and from the RHS of (46), we have $\hat{V}^k - \hat{V}^{k+1} = \boldsymbol{\eta}^{k+1}\Xi\boldsymbol{\eta}^{k+1} \geq 0$ for all k, where $\boldsymbol{\eta}^{k+1}$ can be written as $\boldsymbol{\eta}^{k+1} = \mathcal{B}\bar{\boldsymbol{\eta}}^{k+1}$ with $\mathcal{B} = \mathrm{diag}(I_s, [B_d]^{\mathrm{diag}}_{d\in\mathbb{N}_{D-1}}, I_s, [\rho B_d]^{\mathrm{diag}}_{d\in\mathbb{N}_{D-1}})$ and

$$\bar{\boldsymbol{\eta}}^{k+1} = \Big[[\tilde{y}_d^{k+1}]^\top_{d\in\mathbb{N}_{D-1}}, [x_{d+1}^{k+1} - x_{d+1}^\star]^\top_{d\in\mathbb{N}_{D-1}}, \\ \big[r_d^{k+1} = \frac{1}{\rho}(y_d^{k+1} - y_d^k)\big]^\top_{d\in\mathbb{N}_{D-1}}, [x_{d+1}^{k+1} - x_{d+1}^k]^\top_{d\in\mathbb{N}_{D-1}}\Big]^\top.$$

Because $\bar{\boldsymbol{\eta}}^{k+1}\mathcal{B}^\top\Xi\bar{\boldsymbol{\eta}}^{k+1} \geq 0$ holds for *every solution trajectory* $\bar{\boldsymbol{\eta}}^{k+1}$ generated by hADMM starting from *any initialization* $([x_d^0]_{d\in\mathbb{N}_D}, [y_d^0]_{d\in\mathbb{N}_{D-1}})$ with *any cost function* f_d satisfying Scenario 3, we can argue that $\mathcal{B}^\top\Xi\mathcal{B} \succeq 0$. This can be proved formally by showing that $\bar{\boldsymbol{\eta}}^1$ can be an arbitrary vector, which can be deduced by proving that

$$v = \begin{bmatrix} v_1 \\ v_2 \\ v_3 \\ v_4 \end{bmatrix} = \begin{bmatrix} [y_d^0 - y_d^\star]_{d\in\mathbb{N}_{D-1}} \\ [x_d^0 - x_d^\star]_{d\in\mathbb{N}_D} \\ [r_1]_{d\in\mathbb{N}_{D-1}} \\ [x_d^1 - x_d^0]_{d\in\mathbb{N}_D} \end{bmatrix}$$

can take any value. We next demonstrate this fact for the special case of quadratic functions for simplicity.

Consider quadratic forms $f_d = \frac{1}{2}x_d^\top Q_d x_d + q_d^\top x_d$, where $Q_d \succ 0$. From the first order optimality condition

$$0 = Q_1 x_1^* + q_1 + A_1^\top y_1^* \tag{57a}$$
$$0 = Q_D x_D^* + q_D + B_{D-1}^\top y_{D-1}^*, \tag{57b}$$
$$0 = Q_d x_d^* + q_d + A_d^\top y_d^* + B_{d-1}^\top y_{d-1}^*, \tag{57c}$$
$$0 = A_d x_d^* + B_d x_{d+1}^\star - c_d,\ d = 1\ldots, D-1. \tag{57d}$$

It then yields $-G\begin{bmatrix} [x_d^\star]_{d\in\mathbb{N}_D} \\ [y_d^\star]_{d\in\mathbb{N}_{D-1}} \end{bmatrix} = \begin{bmatrix} [q_d]_{d\in\mathbb{N}_D} \\ [c_d]_{d\in\mathbb{N}_{D-1}} \end{bmatrix} = \bar{q}$ or

$$G\begin{bmatrix} v_2 \\ v_1 \end{bmatrix} = G\begin{bmatrix} x^0 \\ y^0 \end{bmatrix} + \bar{q}, \tag{58}$$

where $\bar{q} = [[q_{d+1}]^\top_{d\in\mathbb{N}_{D-1}}, [c_d]^\top_{d\in\mathbb{N}_{D-1}}]^\top$, $x^k = [x_d^k]_{d\in\mathbb{N}_D}$, $y^k = [y_d^k]_{d\in\mathbb{N}_{D-1}}$, and KKT matrix

$$G = \begin{bmatrix} [Q_d]^{\mathrm{diag}}_{d\in\mathbb{N}_D} & M^\top \\ M & 0 \end{bmatrix},\ M = \begin{bmatrix} A_1 & B_1 & & & \\ 0 & A_2 & B_2 & & \\ \vdots & & \ddots & \ddots & \\ 0 & \ldots & 0 & A_{D-1} & B_{D-1} \end{bmatrix}.$$

On the other hand, from the proof of Theorem 1

$$\begin{aligned} 0 &= Q_1 x_1^1 + q_1 + A_1^\top(y_1^0 + \rho(A_1 x_1^1 + B_1 x_2^0 - c_1)), \\ &= (Q_1 + \rho A_1^\top A_1)x_1^1 + q_1 + A_1^\top y_1^0 + \rho A_1^\top B_1 x_2^0 \underbrace{-\rho A_1^\top c_1}_{h_1}, \end{aligned}$$

$$\begin{aligned} 0 &= Q_D x_D^1 + q_D \\ &\quad + B_{D-1}^\top(y_{D-1}^0 + \rho(A_{D-1}x_{D-1}^1 + B_{D-1}x_D^1 - c_{D-1})), \\ &= (Q_D + \rho B_{D-1}^\top B_{D-1})x_D^1 + \rho B_{D-1}^\top A_{D-1}x_{D-1}^1 + q_D \\ &\quad + B_{D-1}^\top y_{D-1}^0 \underbrace{-\rho B_{D-1}^\top c_{D-1}}_{h_{D-1}}, \end{aligned}$$

$$\begin{aligned} 0 &= Q_d x_d^1 + q_d + A_d^\top(y_d^0 + \rho(A_d x_d^1 + B_d x_{d+1}^0 - c_d)) \\ &\quad + B_{d-1}^\top(y_{d-1}^0 + \rho(A_{d-1}x_{d-1}^1 + B_{d-1}x_d^1 - c_{d-1})), \\ &= (Q_d + \rho A_d^\top A_d + \rho B_{d-1}^\top B_{d-1})x_d^1 + \rho B_{d-1}^\top A_{d-1}x_{d-1}^1 + q_d \\ &\quad + A_d^\top y_d^0 + B_{d-1}^\top y_{d-1}^0 + \rho A_d^\top B_d x_{d+1}^0 \underbrace{-\rho A_d^\top c_d - \rho B_{d-1}^\top c_{d-1}}_{h_d}. \end{aligned}$$

Then $-Hx^1 = h + q + M^\top y^0 + Nx^0$, or

$$Hv_4 = h + q + (H+N)x^0 + M^\top y^0, \tag{59}$$

where $q = [q_d]_{d\in\mathbb{N}_D}$, $h = [h_d]_{d\in\mathbb{N}_D}$, and

$$\begin{aligned} H &= \mathrm{tridiag}\big(\{0,\ldots,0\}, \big\{Q_d + \rho(A_d^\top A_d + B_{d-1}^\top B_{d-1})\big\}_{d\in\mathbb{N}_D}, \\ &\qquad \big\{\rho B_d^\top A_d\big\}_{d\in\mathbb{N}_{D-1}}\big) \\ N &= \mathrm{tridiag}\big(\big\{\rho A_1^\top B_1, \ldots, \rho A_{D-1}^\top B_{D-1}, 0\big\}, \{0,\ldots,0\}, \\ &\qquad \{0,\ldots,0\}\big). \end{aligned}$$

In addition, by letting $c = [c_d]_{d\in\mathbb{N}_{D-1}}$, we have

$$v_3 = Mx^1 - c = -MH^{-1}(h + q + M^\top y^0 + Nx^0) - c. \tag{60}$$

From (58), (59), and (60), since M has full row rank, G and H are non-singular. Then, we can select x^0, y^0, $Q_d \succ 0$, and q_d such that the vectors v_1, v_2, v_3 and v_4 can take any values. Because $\mathcal{B}$ has full row rank and Ξ is symmetric, $\Xi \succeq 0$.

Now that $\Xi \succeq 0$, $\Xi \succ 0$ is equivalent to that the equation $(\boldsymbol{\eta}^{k+1})^\top\Xi\boldsymbol{\eta}^{k+1} = 0$ has only one solution at the origin. The RHS of (46) equals zero if and only if

$$\begin{cases} \tilde{x}_{d+1}^{k+1} = 0, & d = 1,\ldots,D-1, \\ \tilde{x}_{d+1}^{k+1} - \frac{1}{\rho}\delta_{d+1}^{k+1} = 0, & d = 1,\ldots,D-1, \\ r_d^{k+1} - \frac{1}{\rho}\delta_{d+1}^{k+1} = 0, & d = 1,\ldots,D-1, \\ A_d^\top\tilde{y}_d^{k+1} + B_{d-1}^\top\tilde{y}_{d-1}^{k+1} - A_d^\top\delta_{d+1}^{k+1} = 0, & d = 2,\ldots,D-1, \\ B_{D-1}\tilde{y}_{D-1}^{k+1} = 0, \end{cases}$$

or equivalently $\tilde{x}_{d+1}^{k+1} = 0$, $\delta_{d+1}^{k+1} = 0$, $r_d^{k+1} = 0$, and

$$\begin{cases} A_d^\top\tilde{y}_d^{k+1} + B_{d-1}^\top\tilde{y}_{d-1}^{k+1} = 0, \\ B_{D-1}^\top\tilde{y}_{D-1}^{k+1} = 0, \end{cases} \tag{61}$$

for $d = 2,\ldots,D-1$. In addition, (61) can be written as

$$\begin{bmatrix} B_1^\top & A_2^\top & \ldots & 0 \\ 0 & B_2^\top & \ddots & \vdots \\ \vdots & \ddots & \ddots & A_{D-1}^\top \\ 0 & \ldots & 0 & B_{D-1}^\top \end{bmatrix}[\tilde{y}_d^{k+1}]_{d\in\mathbb{N}_{D-1}} = 0. \tag{62}$$

Since B_d has full row rank, the matrix in the LHS of (62) has full column rank, hence $[\tilde{y}_d^{k+1}]_{d\in\mathbb{N}_{D-1}} = 0$ is the unique solution. Thus, $\Xi \succ 0$.

REFERENCES

[1] S. H. Chan, X. Wang, and O. A. Elgendy, "Plug-and-Play ADMM for Image Restoration: Fixed-Point Convergence and Applications," *IEEE Transactions on Computational Imaging*, vol. 3, no. 1, pp. 84–98, Mar. 2017.

[2] Y. Wang, W. Yin, and J. Zeng, "Global convergence of ADMM in nonconvex nonsmooth optimization," *Journal of Scientific Computing*, vol. 78, pp. 29–63, 2019.

[3] T. Erseghe, "Distributed optimal power flow using ADMM," *IEEE transactions on power systems*, vol. 29, no. 5, pp. 2370–2380, 2014.

[4] O. Shorinwa, T. Halsted, J. Yu, and M. Schwager, "Distributed optimization methods for multi-robot systems: Part 2—a survey," *IEEE Robotics & Automation Magazine*, vol. 31, no. 3, pp. 154–169, 2024.

[5] B. D. Biswas, M. S. Hasan, and S. Kamalasadan, "Decentralized distributed convex optimal power flow model for power distribution system based on alternating direction method of multipliers," *IEEE Transactions on Industry Applications*, vol. 59, no. 1, pp. 627–640, 2022.

[6] B. Nguyen, T. Nghiem, L. Nguyen, N. Nguyen, H. La, M. Sookhak, and T. Nguyen, "Distributed formation trajectory planning for multi-vehicle systems," in *Proceedings of the American Control Conference*, 2023.

[7] T. S. Reddy, D. Dhanush, T. Krithin, and S. Jayan, "Supply chain logistics with hybrid optimization using ADMM and vehicle routing problem," in *2024 First International Conference on Innovations in Communications, Electrical and Computer Engineering (ICICEC)*. IEEE, 2024, pp. 1–7.

[8] Y. Noah and N. Shlezinger, "Distributed learn-to-optimize: Limited communications optimization over networks via deep unfolded distributed ADMM," *IEEE Transactions on Mobile Computing*, 2024.

[9] S.-C. Tsai, Y.-H. Tseng, and T.-H. Chang, "Communication-Efficient Distributed Demand Response: A Randomized ADMM Approach," *IEEE Transactions on Smart Grid*, vol. 8, no. 3, pp. 1085–1095, May 2017.

[10] P. Siano, "Demand response and smart grids—A survey," *Renewable and Sustainable Energy Reviews*, vol. 30, pp. 461–478, Feb. 2014.

[11] A. Nedic and A. Ozdaglar, "Distributed subgradient methods for multi-agent optimization," *IEEE Transactions on automatic control*, vol. 54, no. 1, pp. 48–61, 2009.

[12] A. Nedic, A. Ozdaglar, and P. A. Parrilo, "Constrained consensus and optimization in multi-agent networks," *IEEE Transactions on Automatic Control*, vol. 55, no. 4, pp. 922–938, 2010.

[13] J. F. C. Mota, J. M. F. Xavier, P. M. Q. Aguiar, and M. Puschel, "D-ADMM: A Communication-Efficient Distributed Algorithm for Separable Optimization," *IEEE Transactions on Signal Processing*, vol. 61, no. 10, pp. 2718–2723, May 2013.

[14] A. Makhdoumi and A. Ozdaglar, "Convergence rate of distributed ADMM over networks," *IEEE Transactions on Automatic Control*, vol. 62, no. 10, pp. 5082–5095, 2017.

[15] Y. Yang, X. Guan, Q.-S. Jia, L. Yu, B. Xu, and C. J. Spanos, "A survey of ADMM variants for distributed optimization: Problems, algorithms and features," *arXiv preprint arXiv:2208.03700*, 2022.

[16] B. Khaki, C. Chu, and R. Gadh, "A Hierarchical ADMM Based Framework for EV Charging Scheduling," in *2018 IEEE/PES Transmission and Distribution Conference and Exposition (T D)*, Apr. 2018, pp. 1–9.

[17] X. Zhang, D. Biagioni, P. Graf, and J. King, "Cooperative Load Scheduling for Multiple Aggregators Using Hierarchical ADMM," in *2020 IEEE Power & Energy Society Innovative Smart Grid Technologies Conference (ISGT)*, Feb. 2020, pp. 1–5.

[18] S. Boyd, N. Parikh, E. Chu, B. Peleato, J. Eckstein *et al.*, "Distributed optimization and statistical learning via the alternating direction method of multipliers," *Foundations and Trends® in Machine learning*, vol. 3, no. 1, pp. 1–122, 2011.

[19] B. He, M. Tao, and X. Yuan, "Alternating Direction Method with Gaussian Back Substitution for Separable Convex Programming," *SIAM Journal on Optimization*, vol. 22, no. 2, pp. 313–340, Jan. 2012.

[20] T. Goldstein, B. O Donoghue, S. Setzer, and R. Baraniuk, "Fast alternating direction optimization methods," *SIAM Journal on Imaging Sciences*, vol. 7, no. 3, pp. 1588–1623, 2014.

[21] W. Deng and W. Yin, "On the Global and Linear Convergence of the Generalized Alternating Direction Method of Multipliers," *Journal of Scientific Computing*, vol. 66, no. 3, pp. 889–916, Mar. 2016.

[22] T. Lin, S. Ma, and S. Zhang, "On the Global Linear Convergence of the ADMM with Multiblock Variables," *SIAM Journal on Optimization*, vol. 25, no. 3, pp. 1478–1497, Jan. 2015.

[23] W. Deng, M.-J. Lai, Z. Peng, and W. Yin, "Parallel multi-block ADMM with o (1/k) convergence," *Journal of Scientific Computing*, vol. 71, pp. 712–736, 2017.

[24] F. Etedadi, S. Kelouwani, K. Agbossou, N. Henao, F. Laurencelle, and S. S. Hosseini, "Hierarchical transactive home energy management system groups coordination through multi-level consensus sharing-based distributed ADMM," *Sustainable Energy, Grids and Networks*, vol. 39, p. 101460, 2024.

[25] R. Mirzaeifard and S. Werner, "Smoothing ADMM for non-convex and non-smooth hierarchical federated learning," in *2025 33rd European Signal Processing Conference (EUSIPCO)*, 2025, pp. 1178–1182.

[26] S. M. Azimi-Abarghouyi, N. Bastianello, K. H. Johansson, and V. Fodor, "Hierarchical Federated ADMM," *IEEE Networking Letters*, vol. 7, no. 1, pp. 11–15, Mar. 2025.

[27] D. Goldfarb and S. Ma, "Fast multiple-splitting algorithms for convex optimization," *SIAM Journal on Optimization*, vol. 22, no. 2, pp. 533–556, 2012.

[28] X. Cao and K. R. Liu, "Dynamic sharing through the ADMM," *IEEE Transactions on Automatic Control*, vol. 65, no. 5, pp. 2215–2222, 2019.

[29] W. Deng and W. Yin, "On the global and linear convergence of the generalized alternating direction method of multipliers," *Journal of Scientific Computing*, vol. 66, pp. 889–916, 2016.

[30] P. Braun, T. Faulwasser, L. Grüne, C. M. Kellett, S. R. Weller, and K. Worthmann, "Hierarchical distributed ADMM for predictive control with applications in power networks," *IFAC Journal of Systems and Control*, vol. 3, pp. 10–22, Mar. 2018.

[31] S. P. Boyd and L. Vandenberghe, *Convex optimization*. Cambridge university press, 2004.

[32] S. Boyd, L. Xiao, and A. Mutapcic, "Subgradient methods," *lecture notes of EE392o, Stanford University, Autumn Quarter*, vol. 2004, no. 01, 2003.